NOISY U-TV | 5%-50% sampling

Held-out PCM evaluation • checkpoint from epoch 5, step 22,500

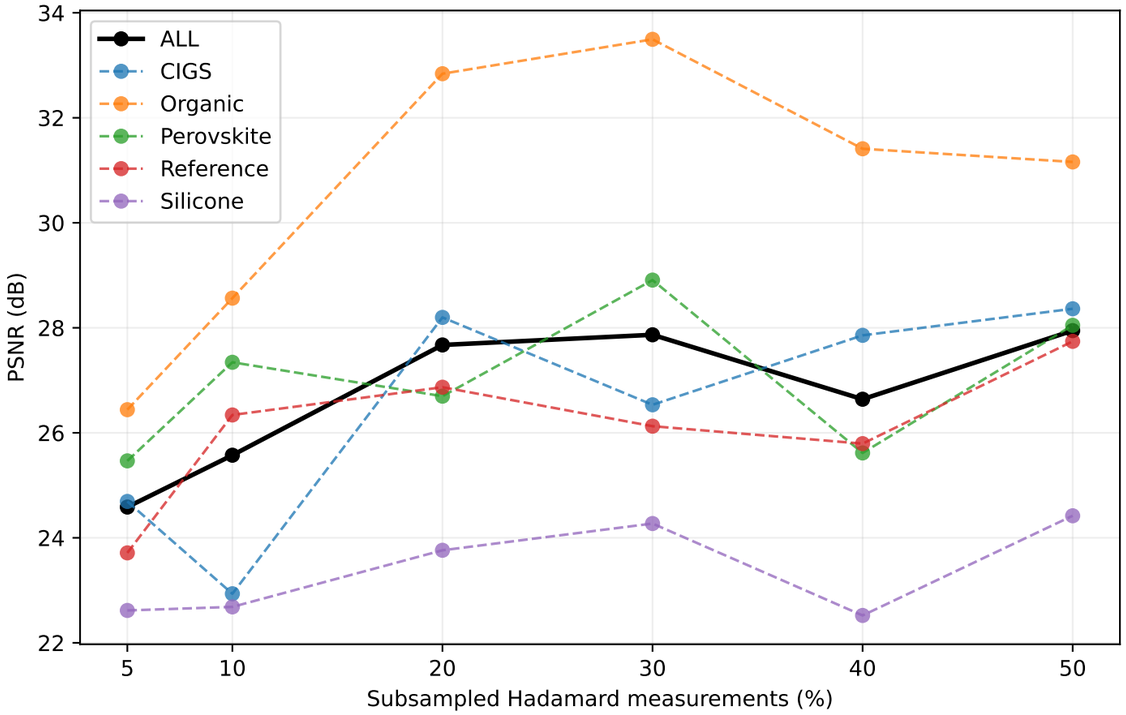
- Model: 4-level U-Net, 128 initial filters, 512 PDHG iterations
- Evaluation GPU: RTX 4070 (training continues separately on RTX 4090)
- Five PCM categories; 10 held-out TEST images per category (50 images total)
- Six sample ratios, coarse J=6; 300 reconstructions
- Fresh, reproducibly seeded noise draw per image and sampling ratio
- Dark / bias / beam sigma sampled from the SAME variable-noise ranges used in training
- One noise realisation per image per ratio; results are descriptive, not final test-set estimates
- Metrics computed by the existing LION compute_metrics implementation (unchanged)
- The learned scale-factor key is shared across all ratios (as in training).
- 5% falls below the fully sampled J6 block; existing sampler selects a subset from J6.

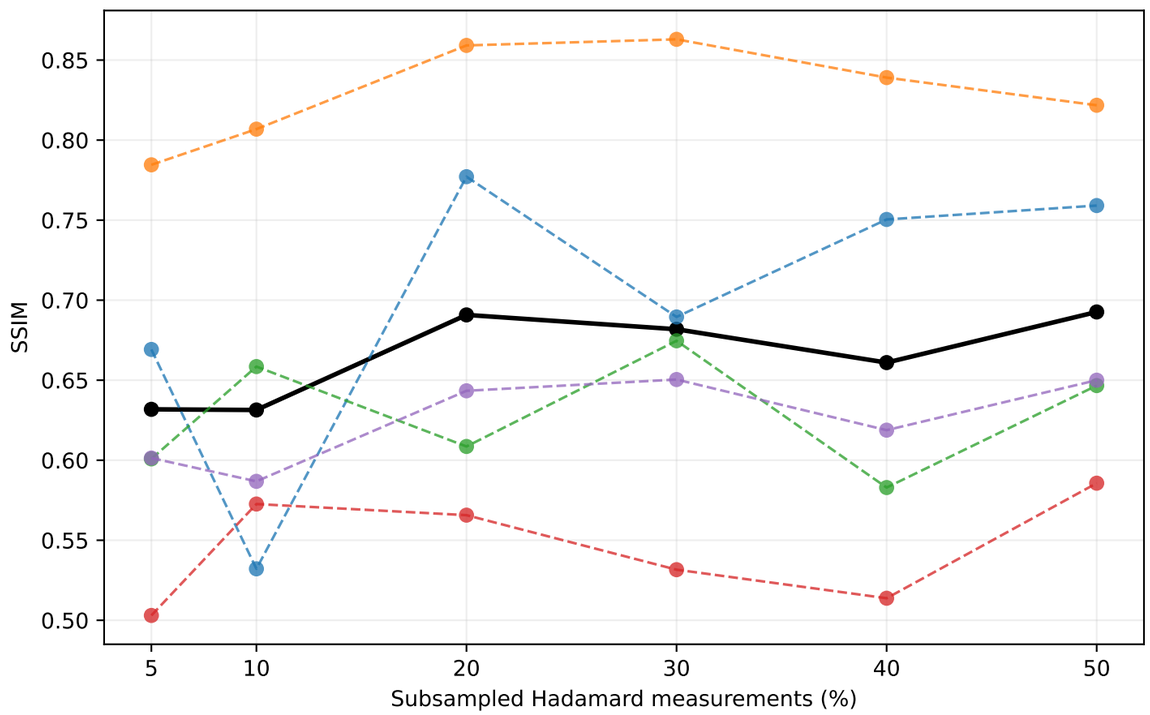
Sampling	Mean MSE	Mean PSNR (dB)	Mean SSIM	Mean HaarPSI
5%	0.00533	24.59	0.632	0.461
10%	0.00439	25.57	0.631	0.509
20%	0.00258	27.67	0.691	0.583
30%	0.00263	27.87	0.682	0.591
40%	0.00335	26.64	0.661	0.535
50%	0.00251	27.95	0.693	0.597

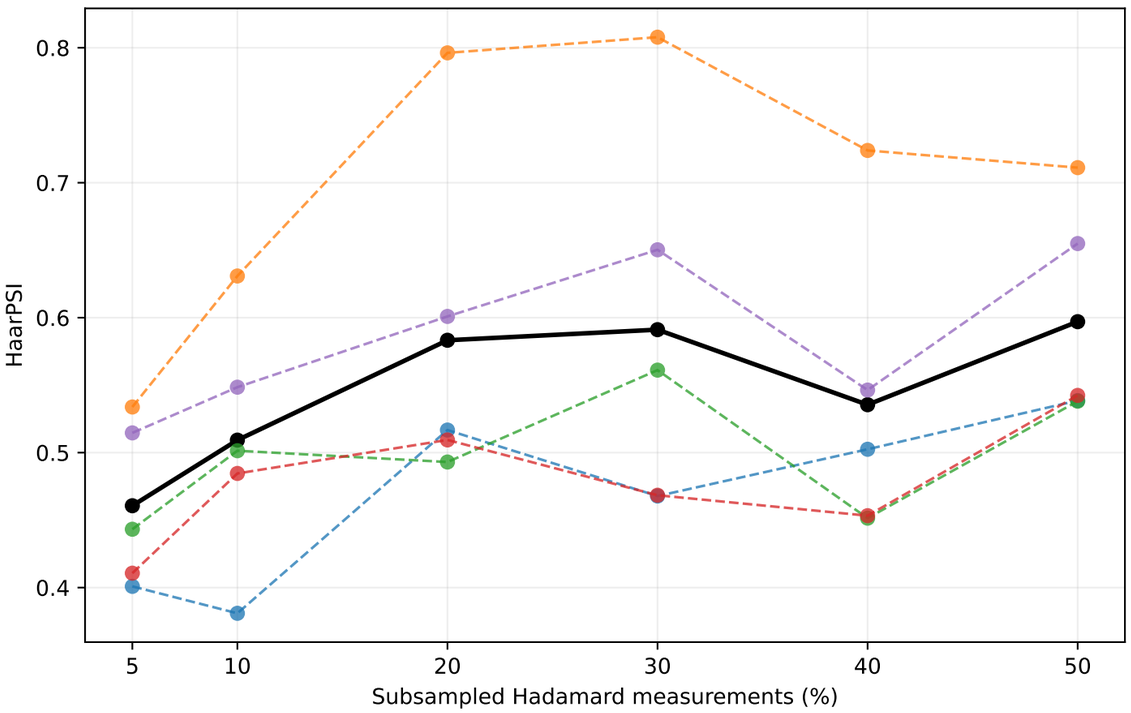
Source: snapshot best_val_loss_s0.20_cJ6.pt; saved training checkpoint is NOT accessed during testing.

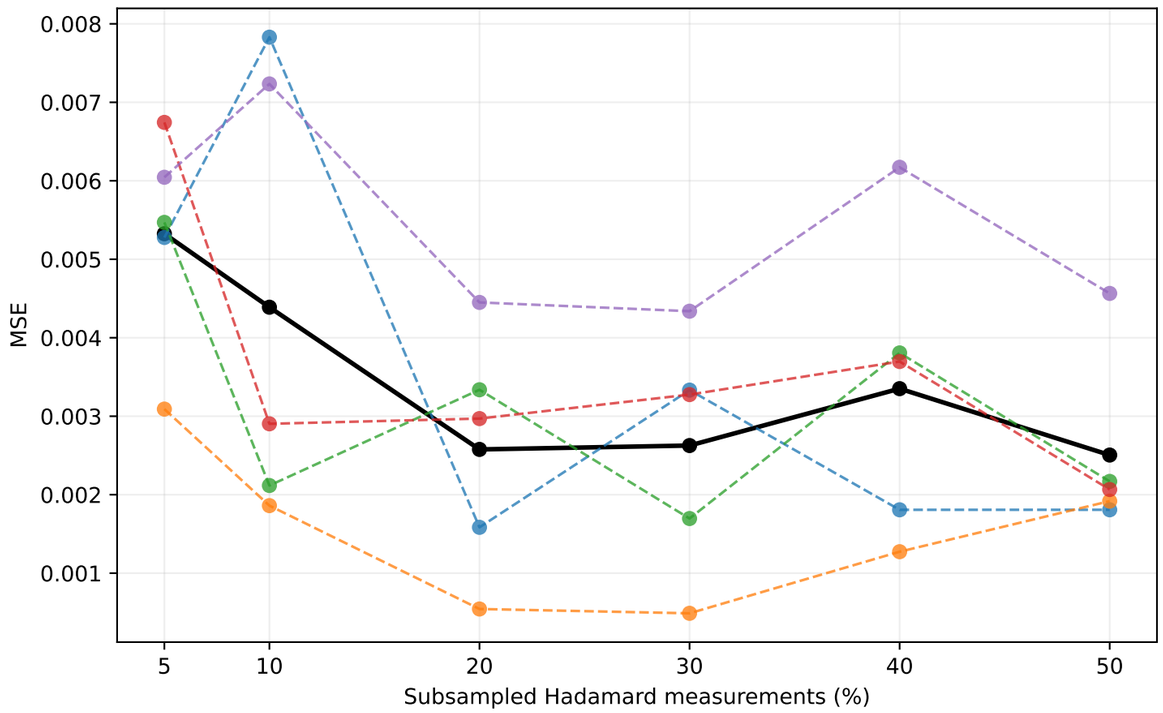
All category averages use equal numbers of samples. See metrics_summary.csv and metrics_per_image.csv.

Sampling ratio response by category





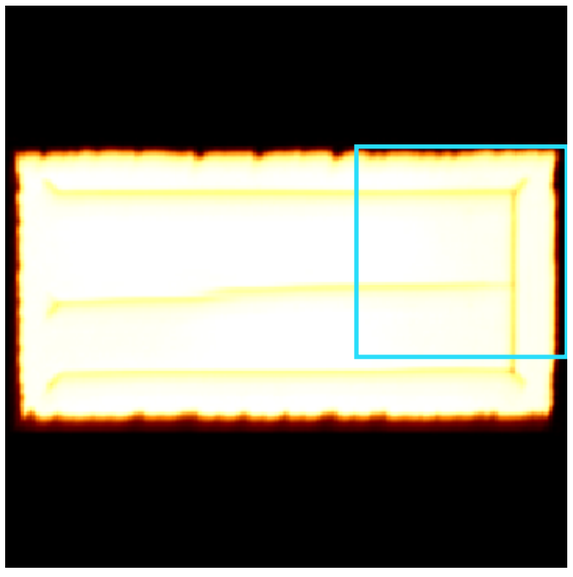




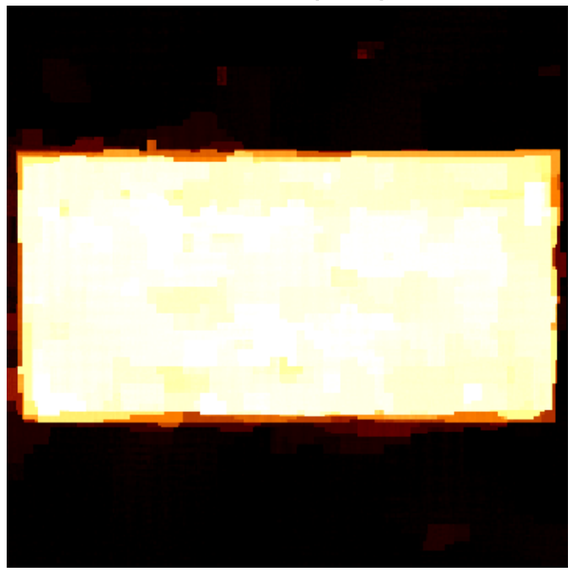
CIGS / image0249 | Full 256x256 field

HELD-OUT TEST SPLIT

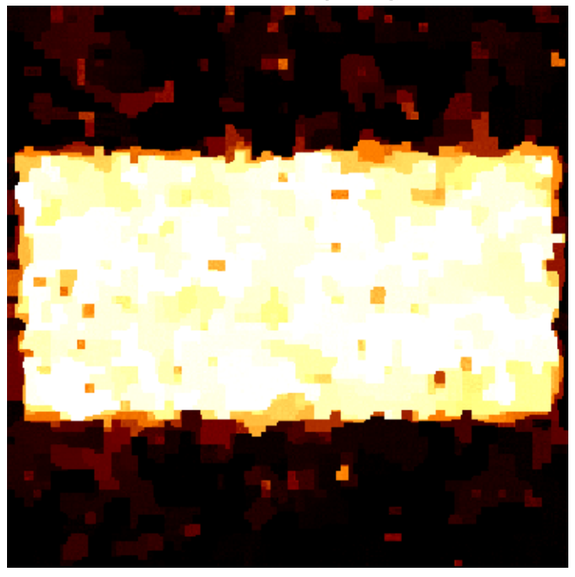
Ground truth



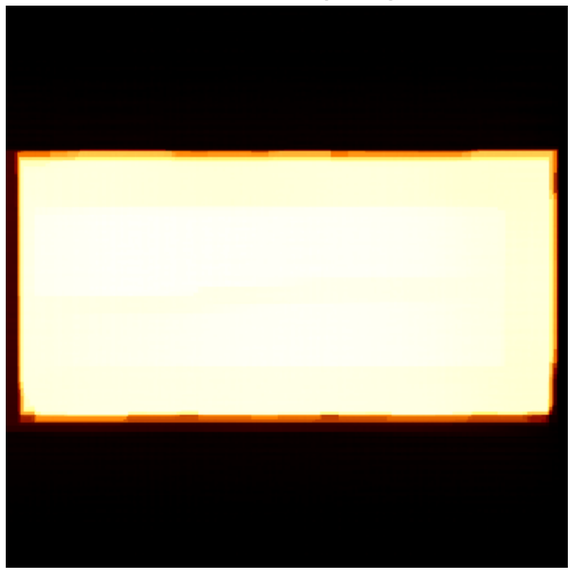
5% sampling



10% sampling



20% sampling



30% sampling



40% sampling



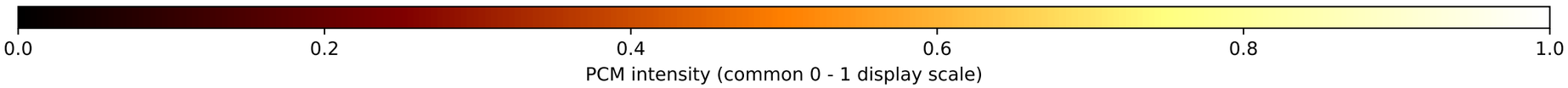
50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI			
5%	24.79 dB	0.627	0.396
10%	18.30 dB	0.259	0.245
20%	27.89 dB	0.767	0.452
30%	28.98 dB	0.793	0.512
40%	28.89 dB	0.774	0.507
50%	22.08 dB	0.478	0.336

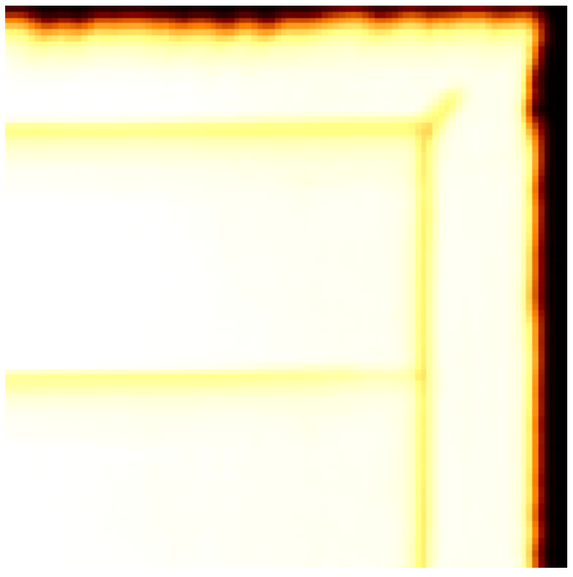
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



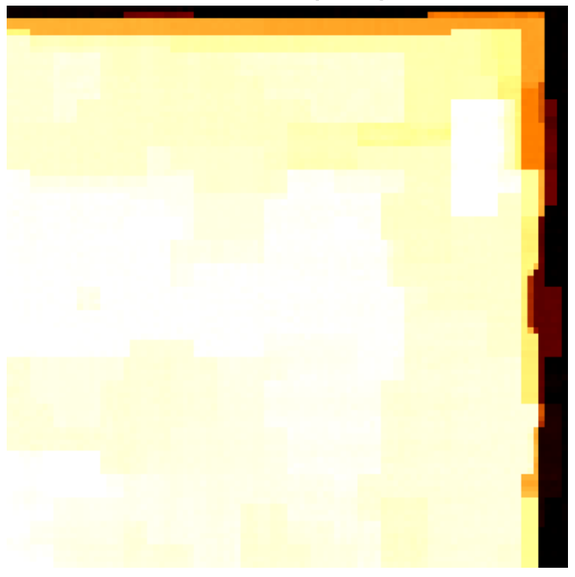
CIGS / image0249 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

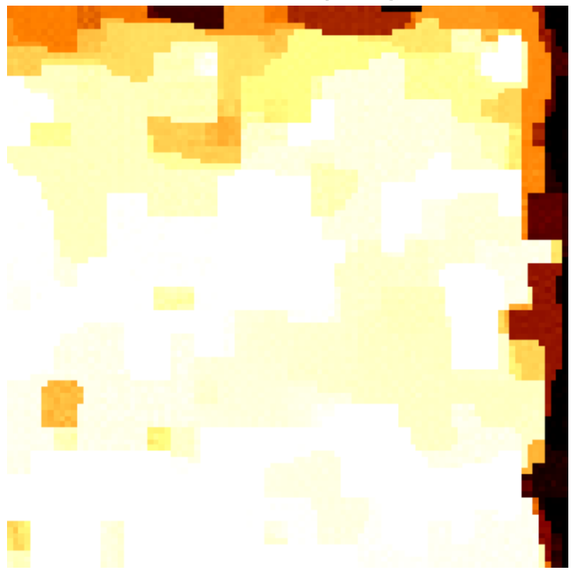
Ground truth



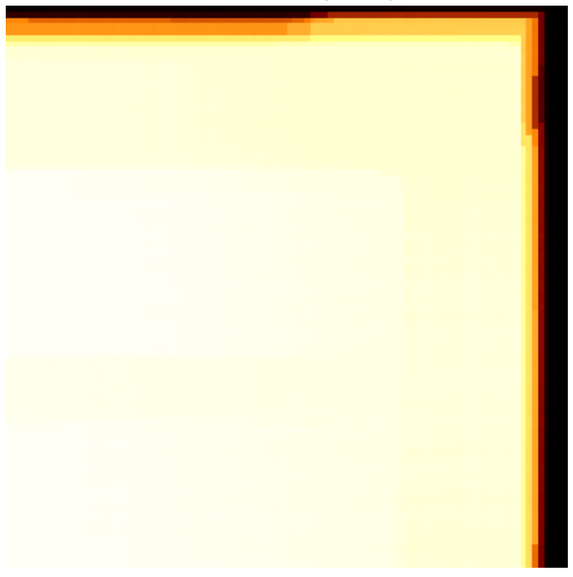
5% sampling



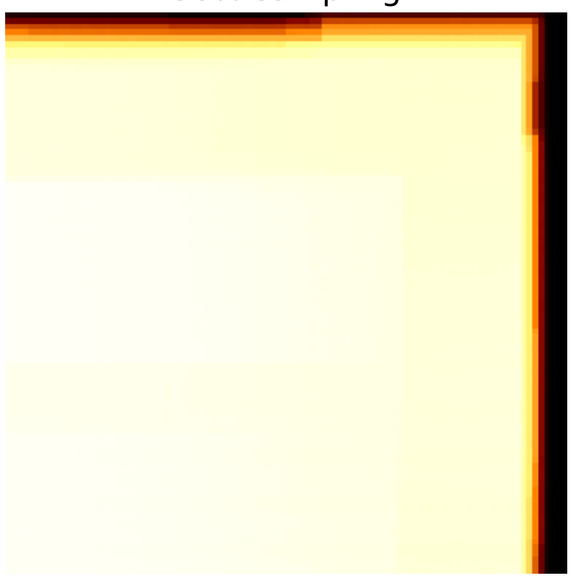
10% sampling



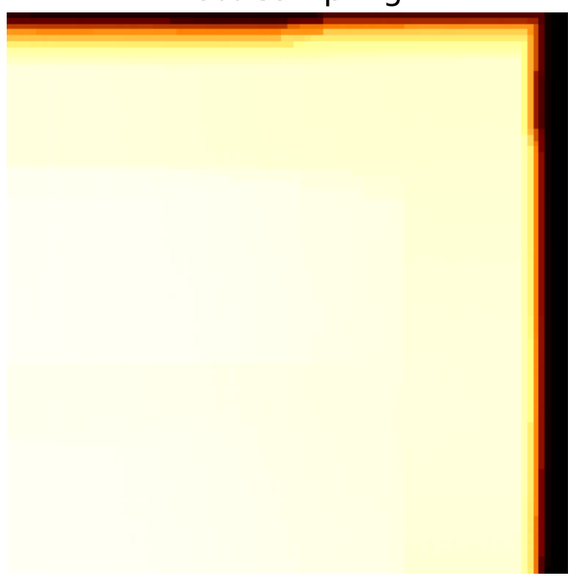
20% sampling



30% sampling



40% sampling



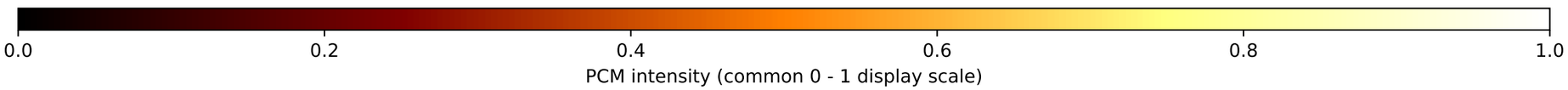
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   24.79 dB  0.627   0.396
10%   18.30 dB  0.259   0.245
20%   27.89 dB  0.767   0.452
30%   28.98 dB  0.793   0.512
40%   28.89 dB  0.774   0.507
50%   22.08 dB  0.478   0.336
```

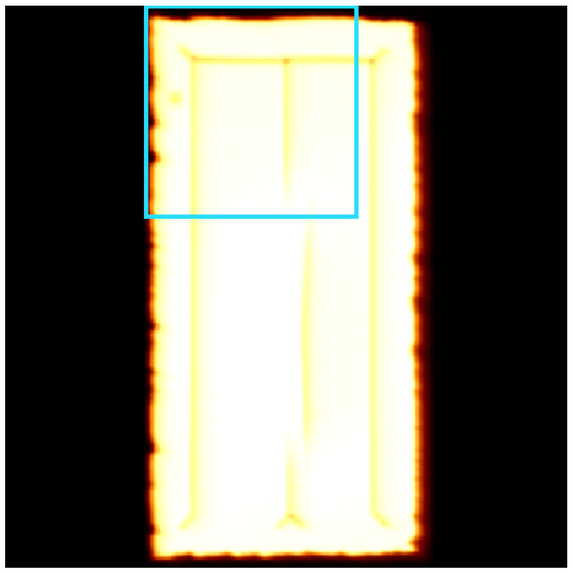
Crop: x=160:256, y=64:160. All panels use the same coordinates and intensity scale.



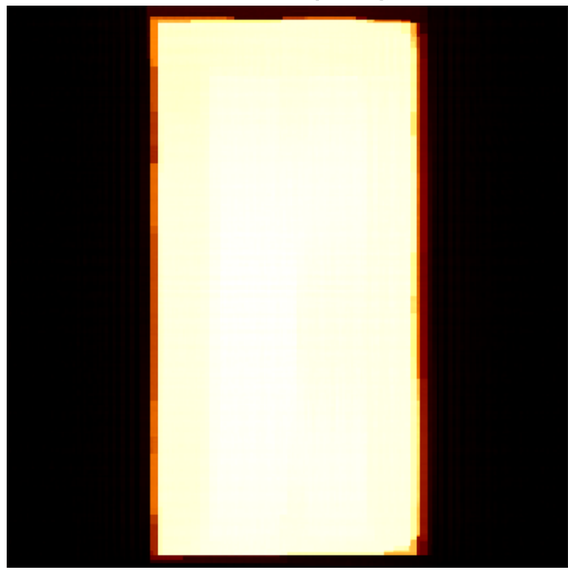
CIGS / image0288 | Full 256x256 field

HELD-OUT TEST SPLIT

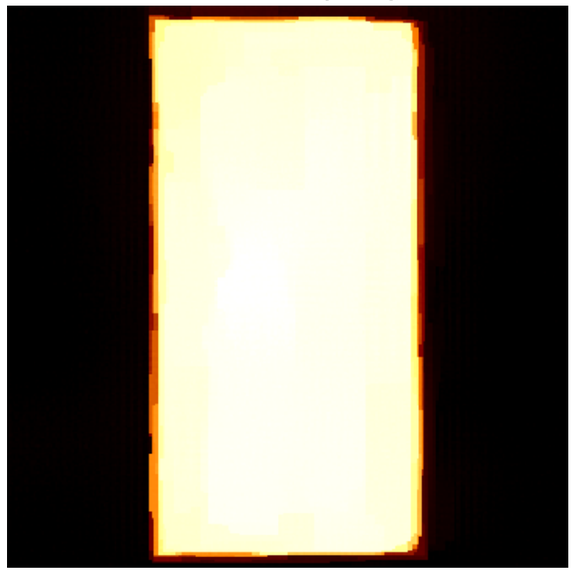
Ground truth



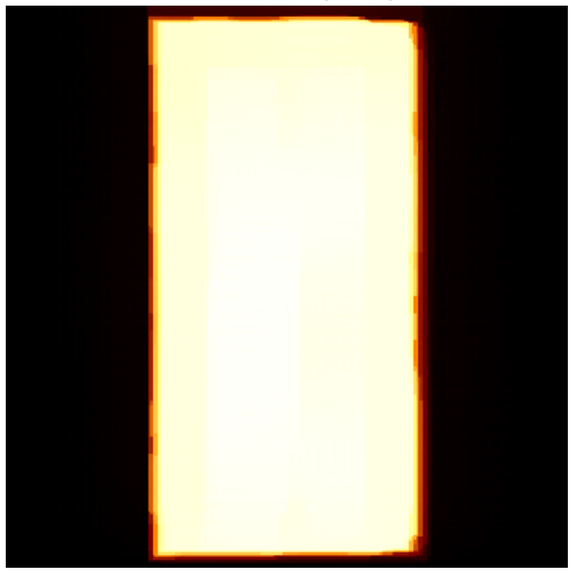
5% sampling



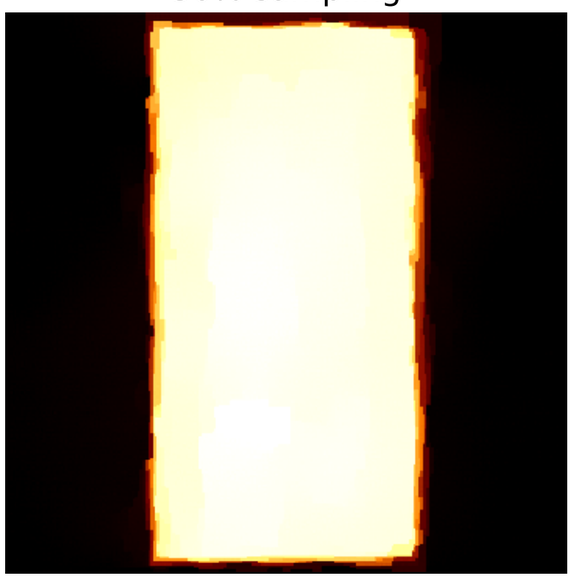
10% sampling



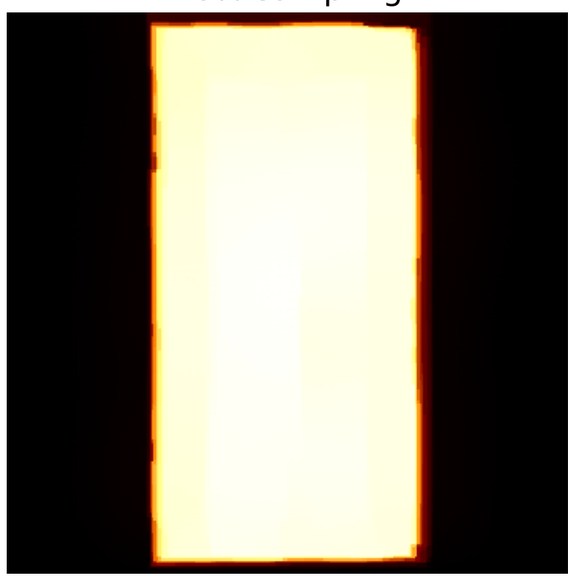
20% sampling



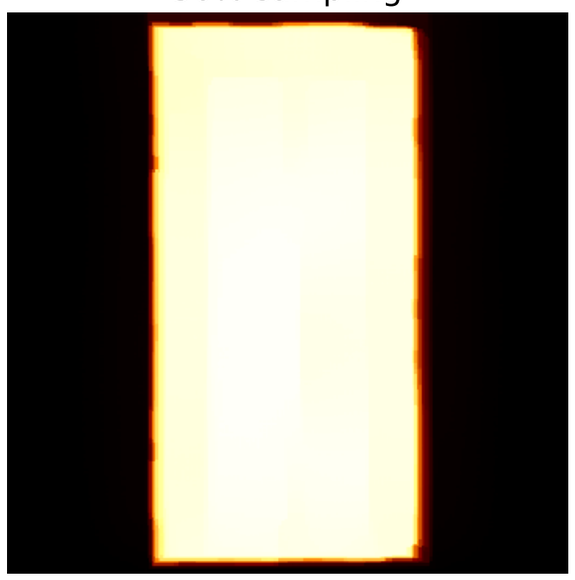
30% sampling



40% sampling



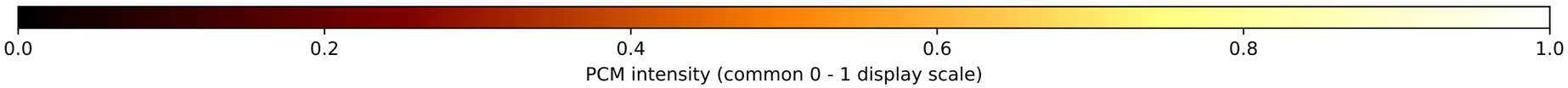
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

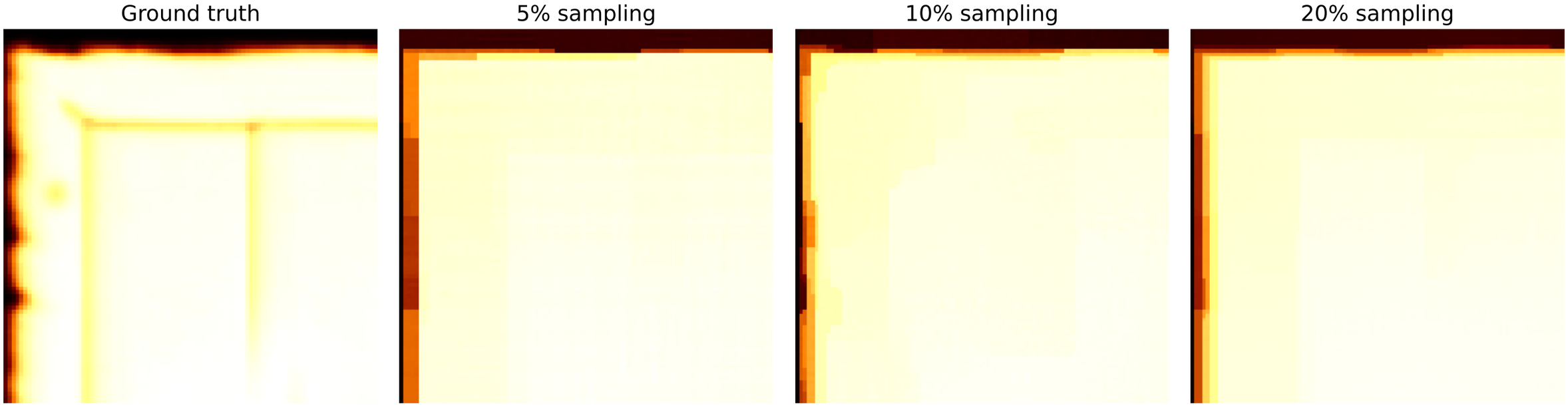
PSNR / SSIM / HaarPSI
 5%   25.86 dB  0.754   0.397
10%   26.80 dB  0.718   0.472
20%   28.31 dB  0.750   0.506
30%   26.83 dB  0.700   0.462
40%   28.84 dB  0.779   0.511
50%   29.19 dB  0.792   0.524
```

Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



CIGS / image0288 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT




```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   25.86 dB  0.754   0.397
10%   26.80 dB  0.718   0.472
20%   28.31 dB  0.750   0.506
30%   26.83 dB  0.700   0.462
40%   28.84 dB  0.779   0.511
50%   29.19 dB  0.792   0.524
```

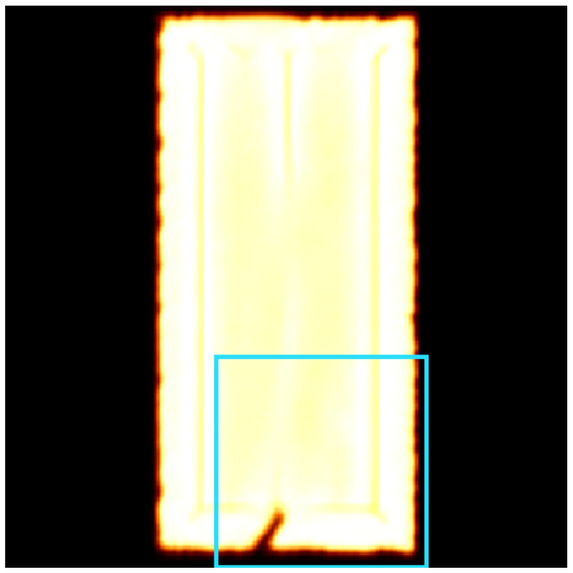
Crop: x=64:160, y=0:96. All panels use the same coordinates and intensity scale.



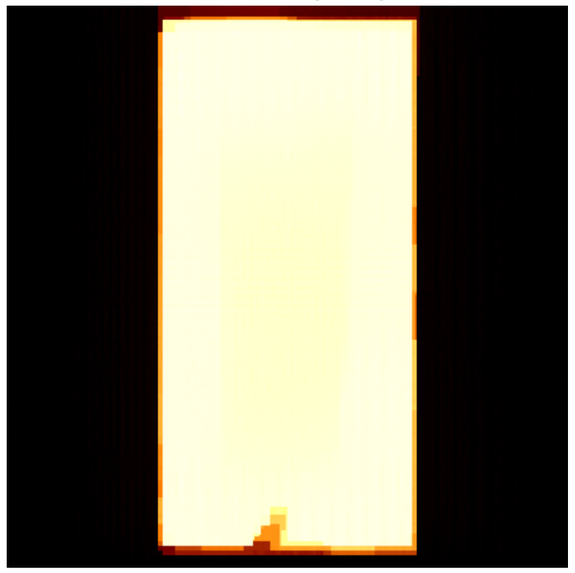
CIGS / image0611 | Full 256x256 field

HELD-OUT TEST SPLIT

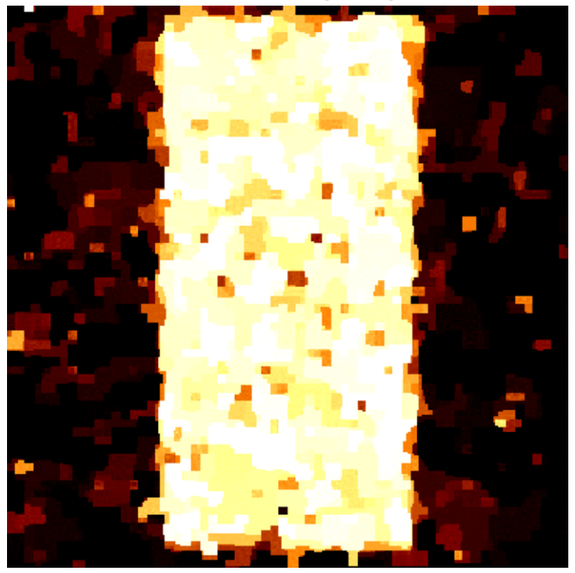
Ground truth



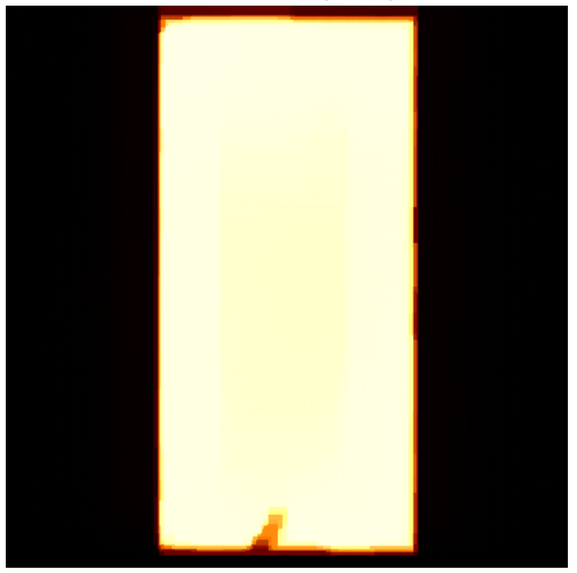
5% sampling



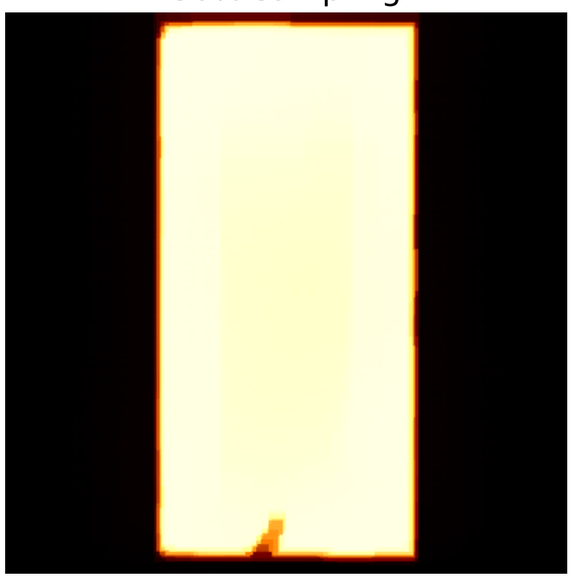
10% sampling



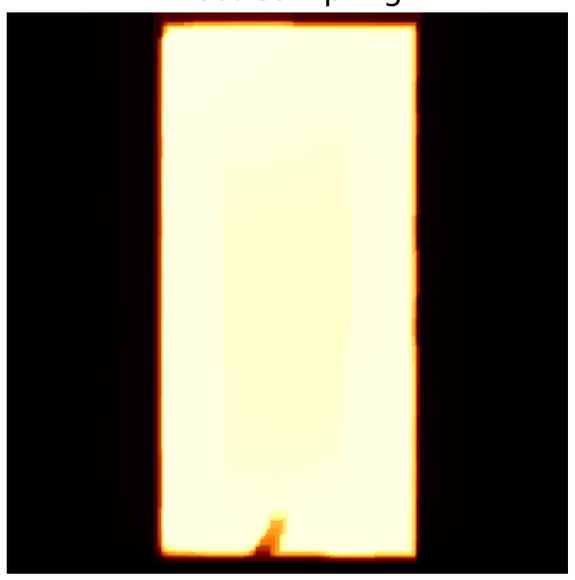
20% sampling



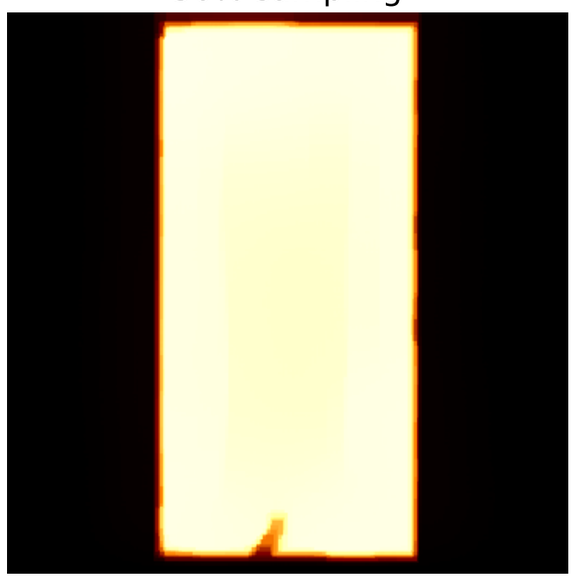
30% sampling



40% sampling



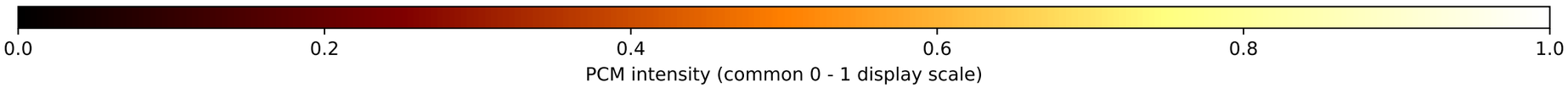
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   26.69 dB  0.748   0.444
10%   15.99 dB  0.182   0.218
20%   28.96 dB  0.780   0.511
30%   29.88 dB  0.795   0.563
40%   30.11 dB  0.803   0.564
50%   30.14 dB  0.795   0.574
```

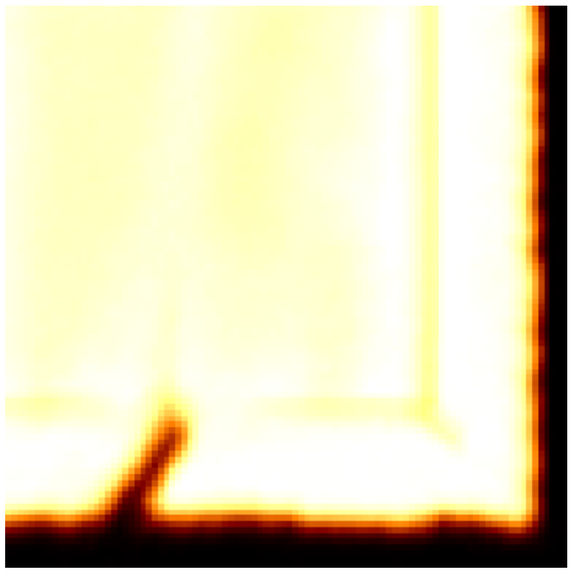
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



CIGS / image0611 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

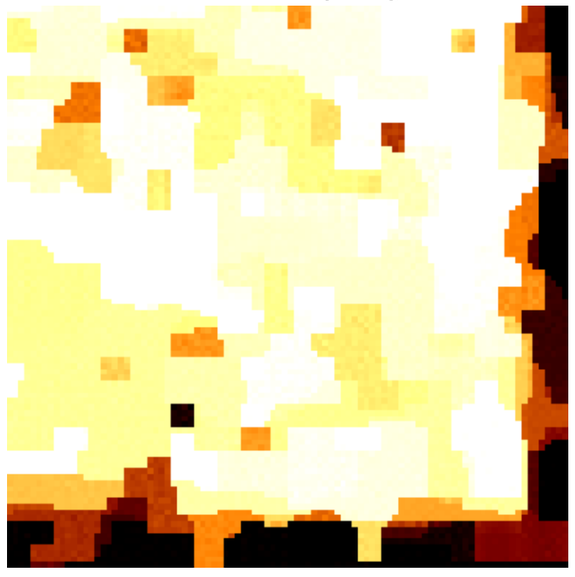
Ground truth



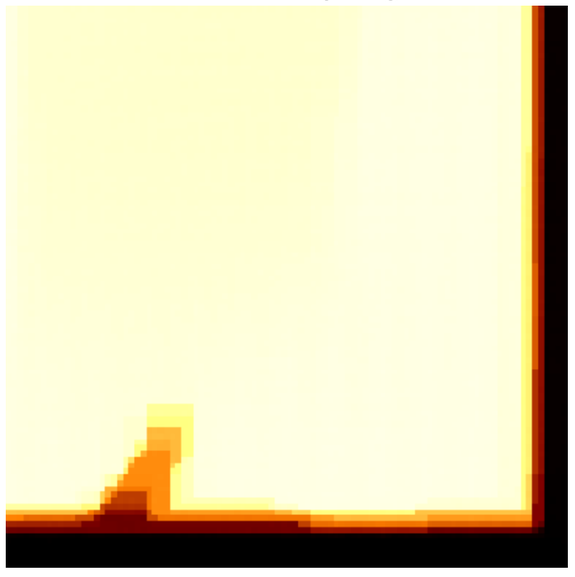
5% sampling



10% sampling



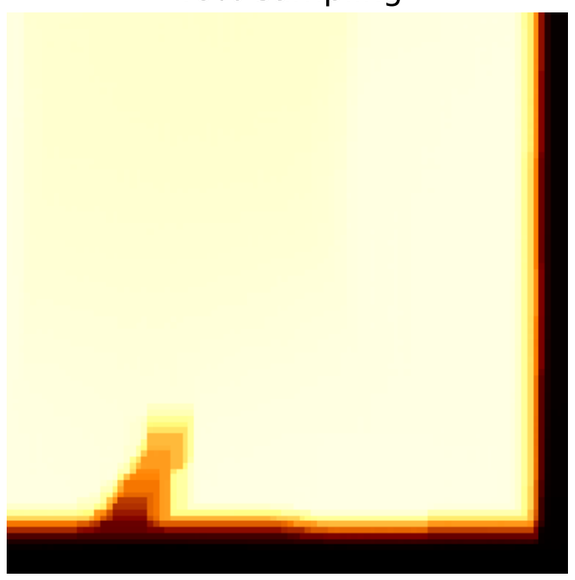
20% sampling



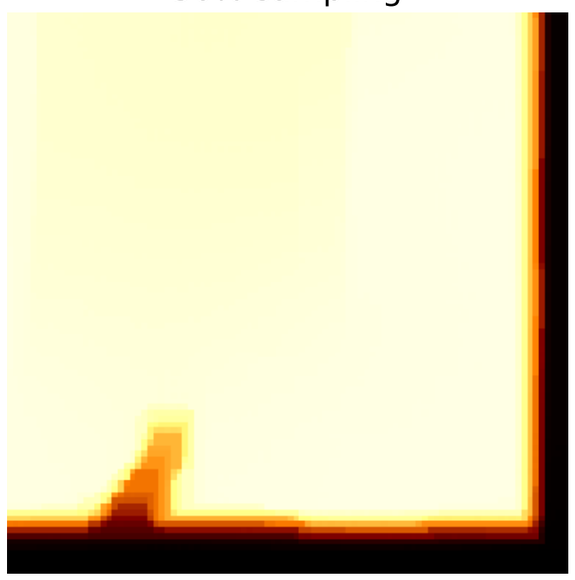
30% sampling



40% sampling



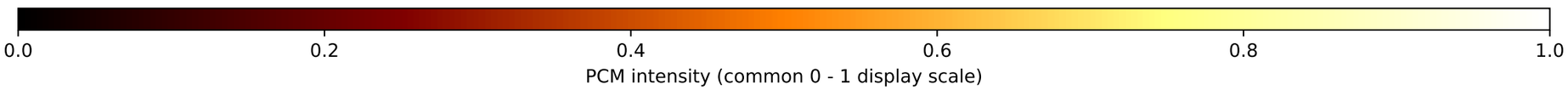
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   26.69 dB  0.748   0.444
10%   15.99 dB  0.182   0.218
20%   28.96 dB  0.780   0.511
30%   29.88 dB  0.795   0.563
40%   30.11 dB  0.803   0.564
50%   30.14 dB  0.795   0.574
```

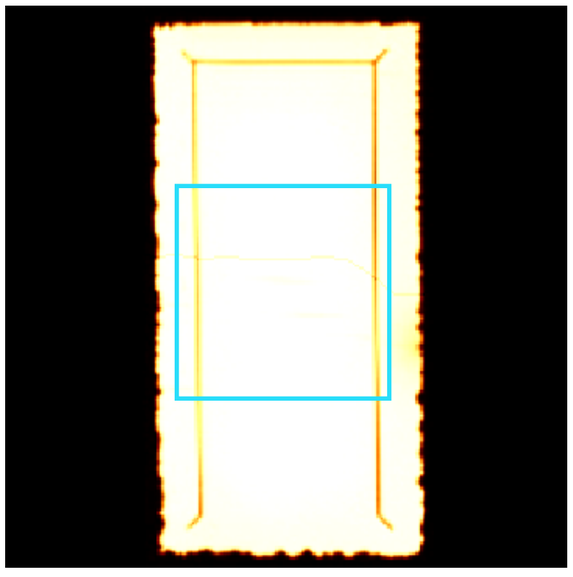
Crop: x=96:192, y=160:256. All panels use the same coordinates and intensity scale.



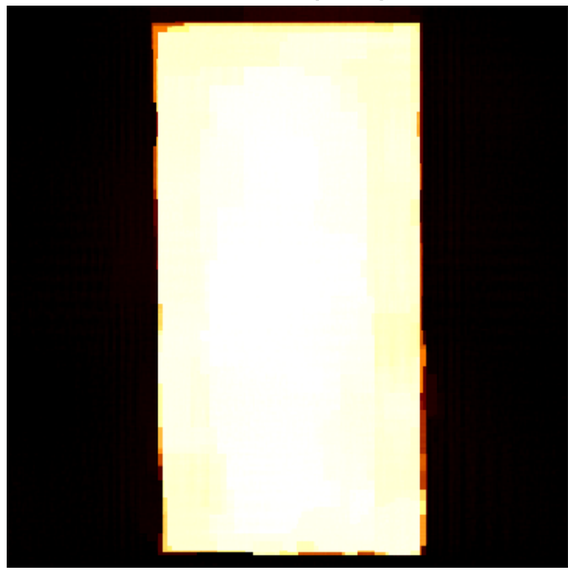
CIGS / image0674 | Full 256x256 field

TRAINING SPLIT - qualitative illustration ONLY (excluded from test metrics)

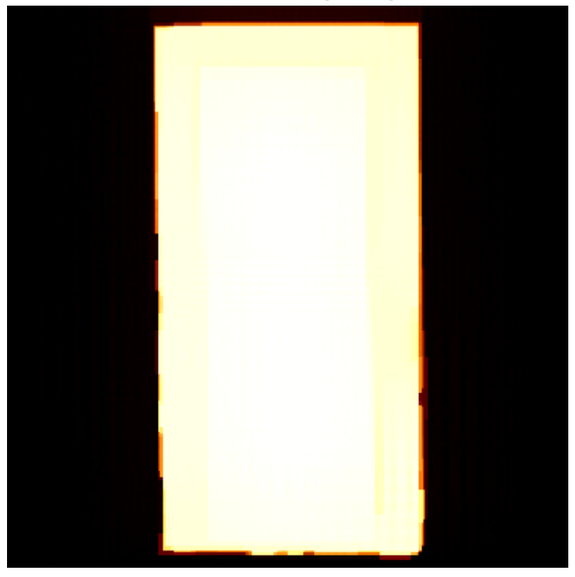
Ground truth



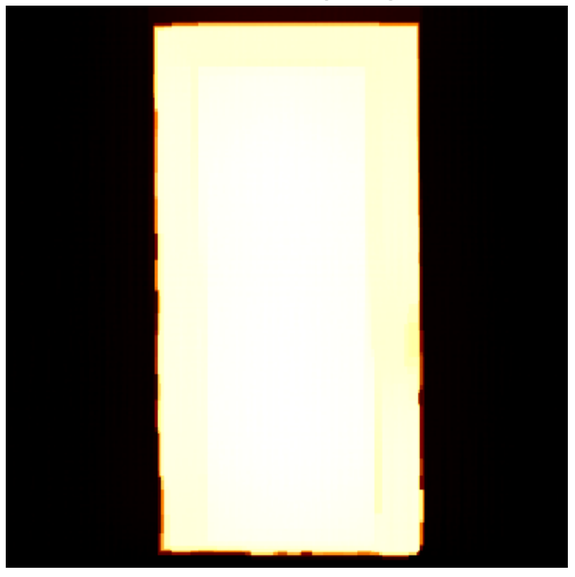
5% sampling



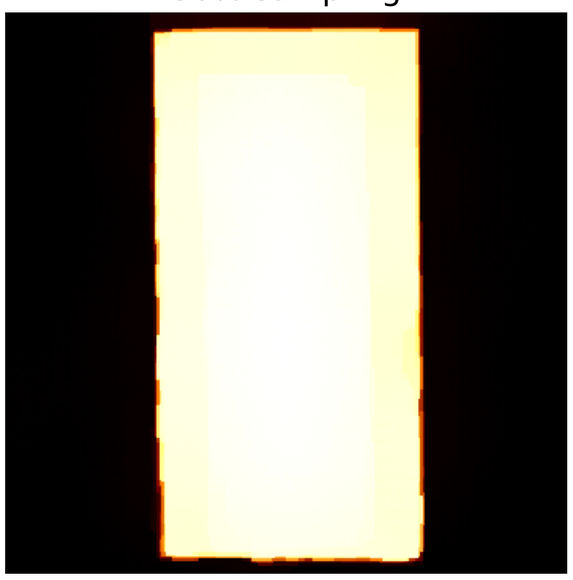
10% sampling



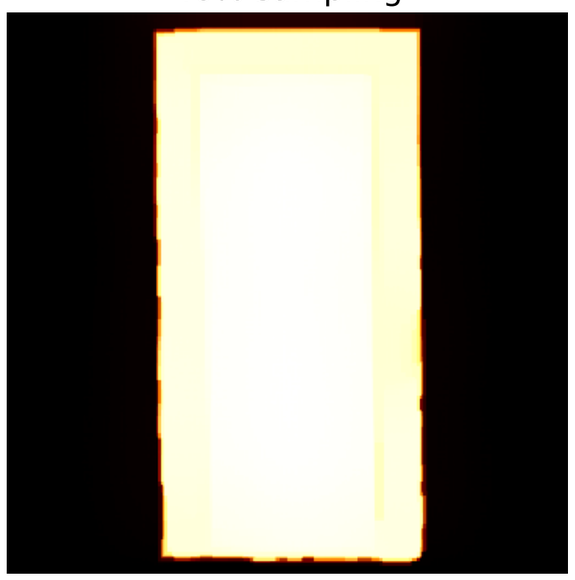
20% sampling



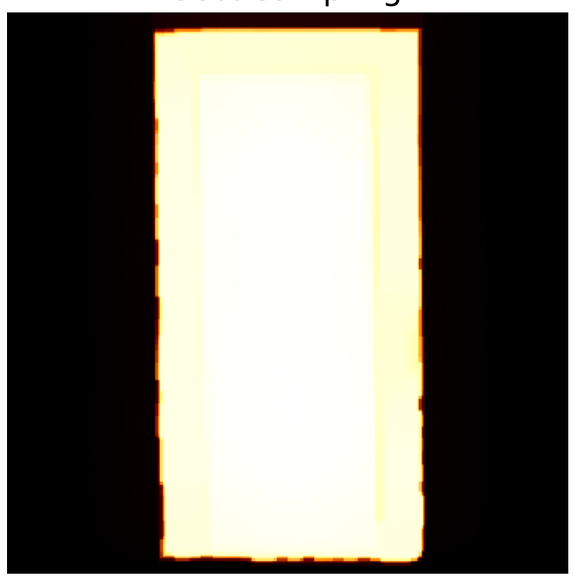
30% sampling



40% sampling



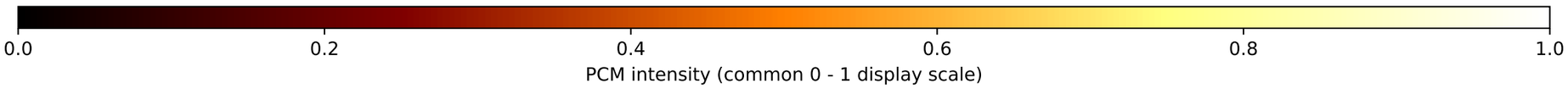
50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

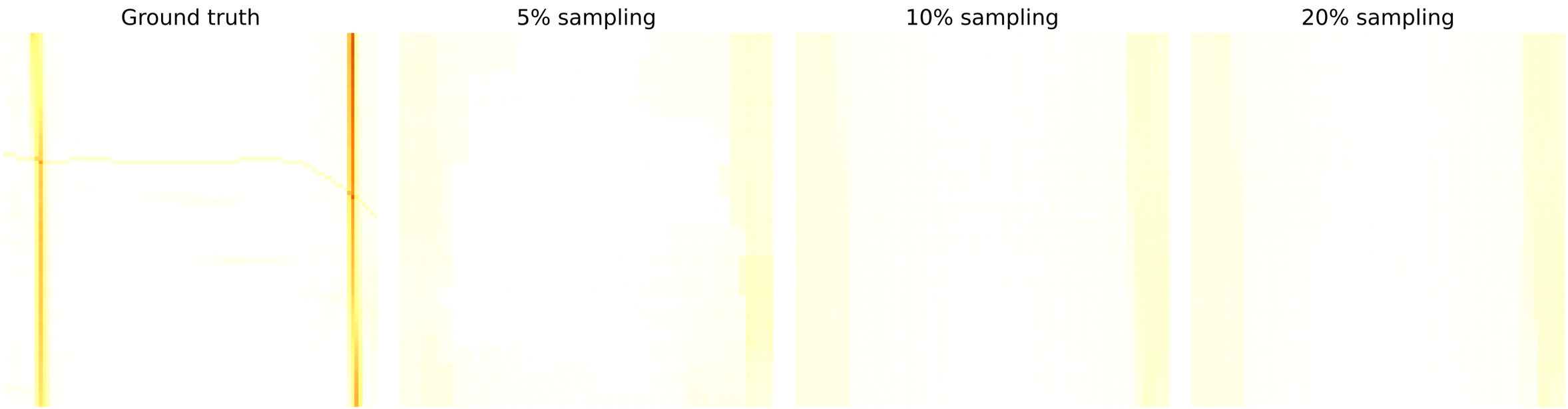
Visual only; not used in
held-out aggregate metrics.

Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



CIGS / image0674 | Thin 1-pixel line detail (manually specified known location)

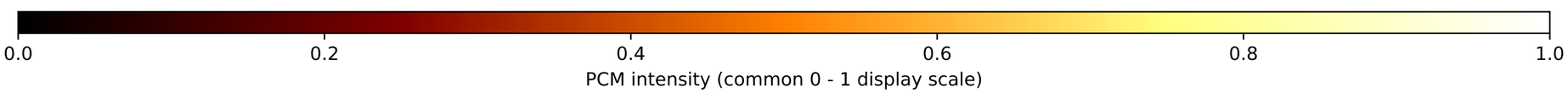
TRAINING SPLIT - qualitative illustration ONLY (excluded from test metrics)




```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

Visual only; not used in
held-out aggregate metrics.
```

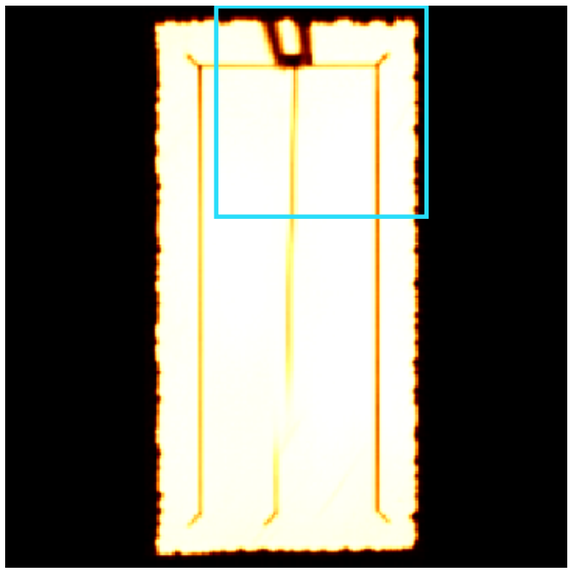
Crop: x=78:175, y=82:179. All panels use the same coordinates and intensity scale.



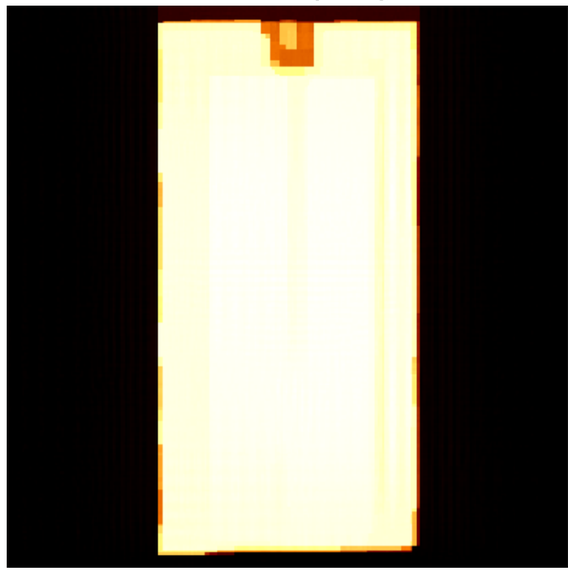
CIGS / image0917 | Full 256x256 field

VALIDATION SPLIT - qualitative illustration ONLY (excluded from test metrics)

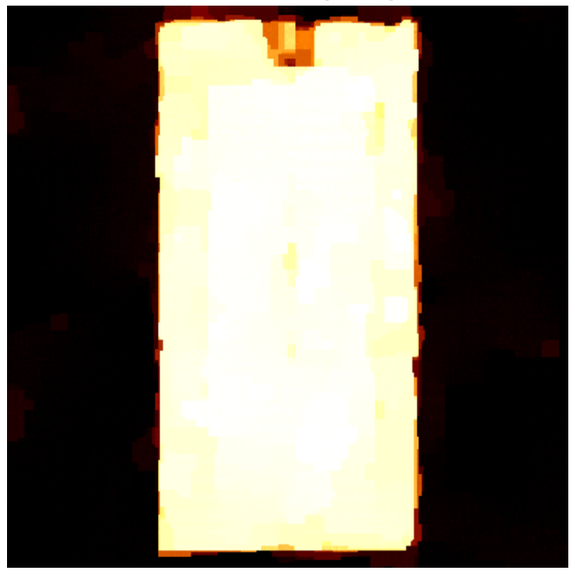
Ground truth



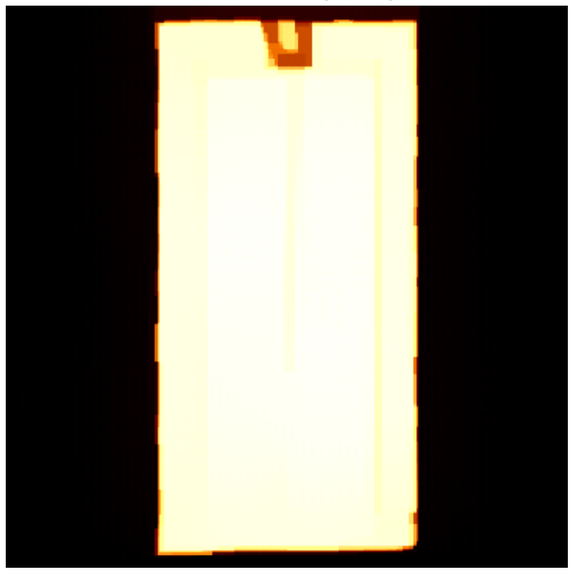
5% sampling



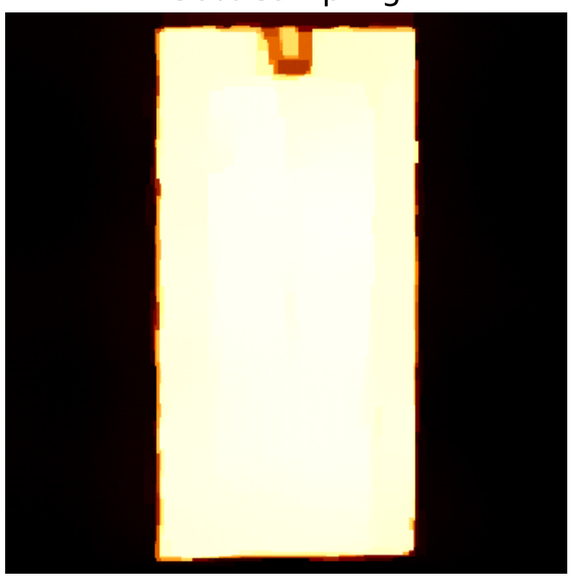
10% sampling



20% sampling



30% sampling



40% sampling



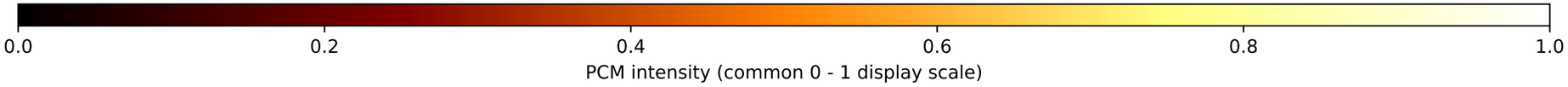
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

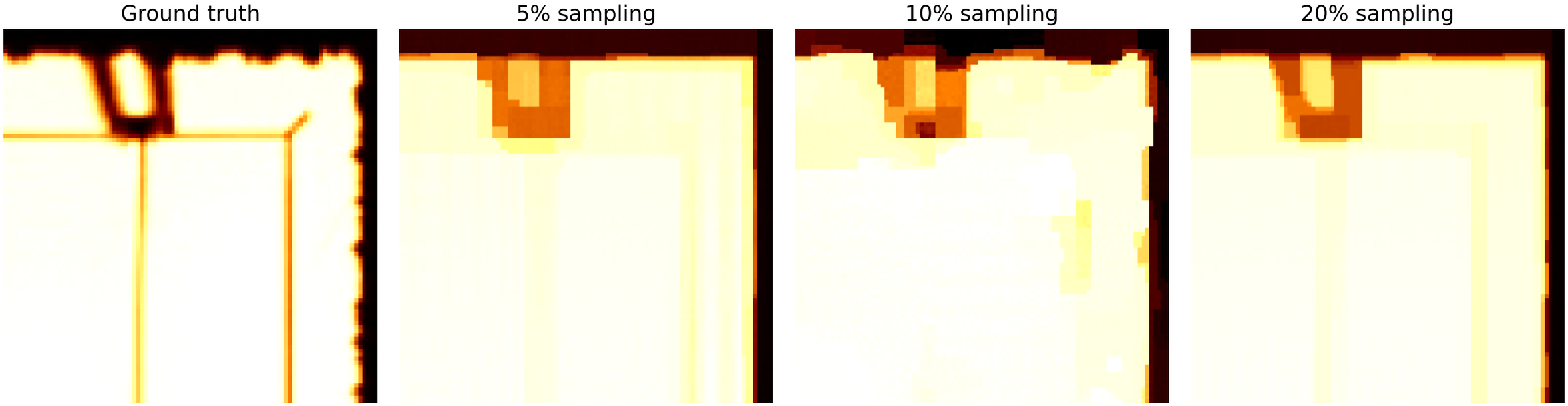
Visual only; not used in
held-out aggregate metrics.
```

Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



CIGS / image0917 | Local detail (automatically selected gradient-rich 96px crop)

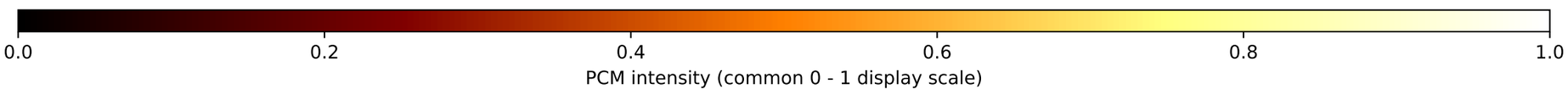
VALIDATION SPLIT - qualitative illustration ONLY (excluded from test metrics)



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

Visual only; not used in
held-out aggregate metrics.

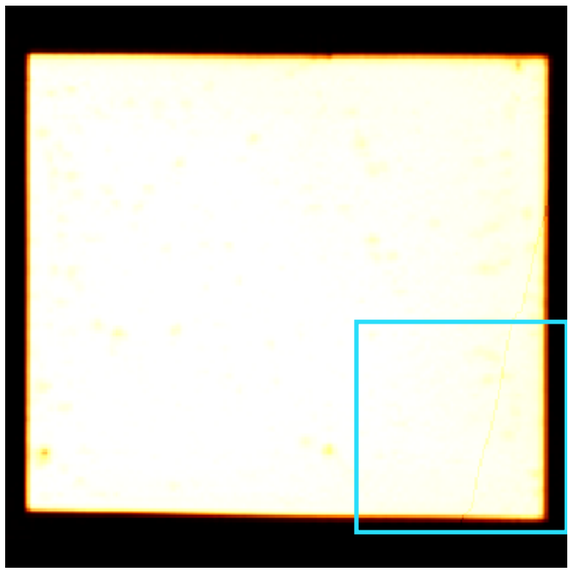
Crop: x=96:192, y=0:96. All panels use the same coordinates and intensity scale.



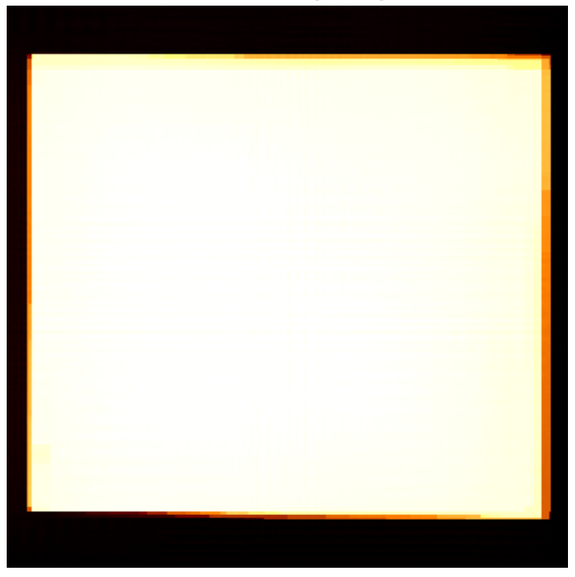
Organic / image0249 | Full 256x256 field

HELD-OUT TEST SPLIT

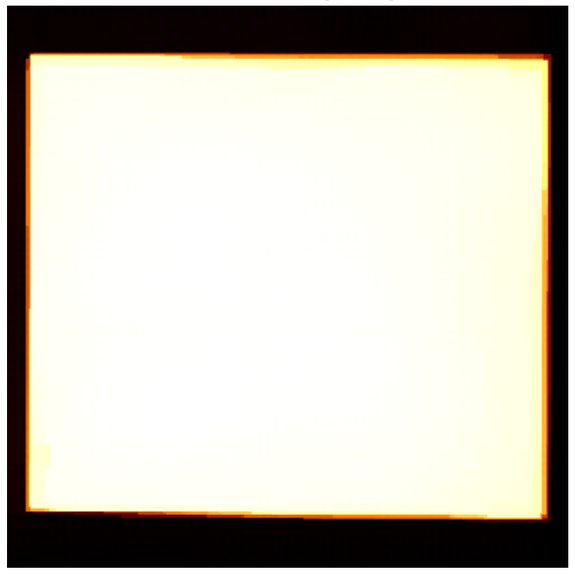
Ground truth



5% sampling



10% sampling



20% sampling



30% sampling



40% sampling



50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI			
5%	27.85 dB	0.828	0.582
10%	31.51 dB	0.839	0.757
20%	33.04 dB	0.847	0.836
30%	33.14 dB	0.848	0.833
40%	35.13 dB	0.860	0.848
50%	36.12 dB	0.869	0.863

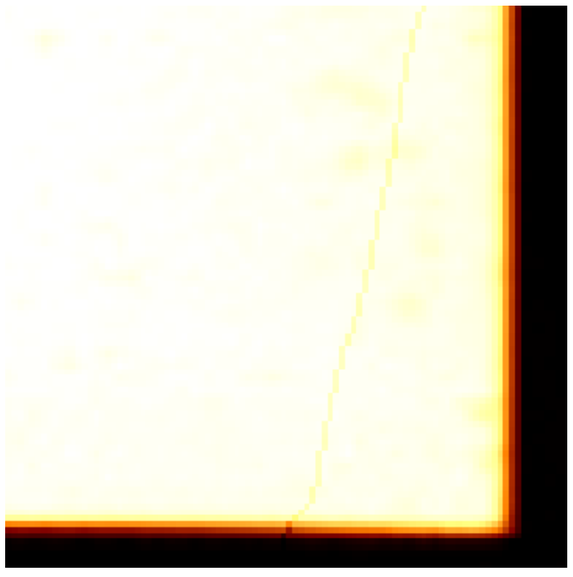
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



Organic / image0249 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

Ground truth



5% sampling



10% sampling



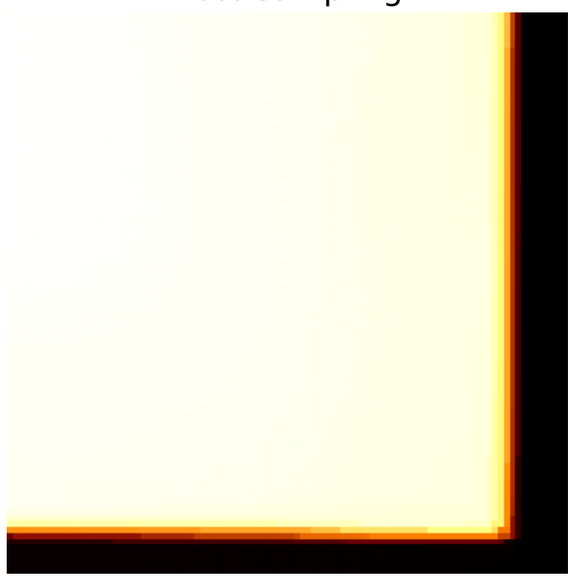
20% sampling



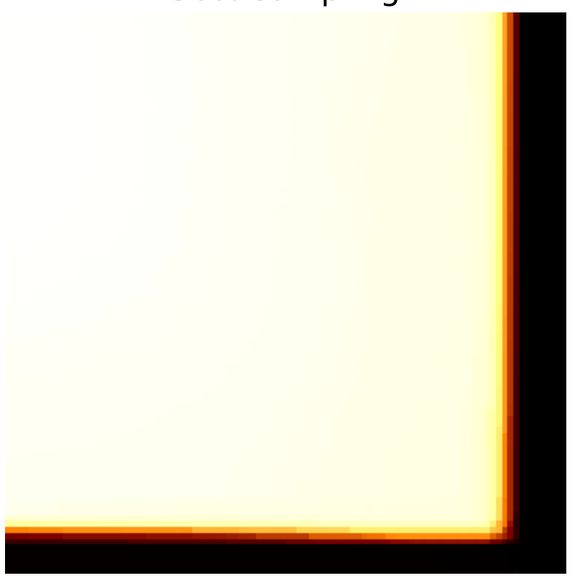
30% sampling



40% sampling



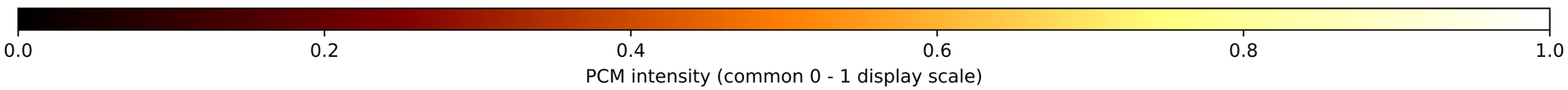
50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI			
5%	27.85 dB	0.828	0.582
10%	31.51 dB	0.839	0.757
20%	33.04 dB	0.847	0.836
30%	33.14 dB	0.848	0.833
40%	35.13 dB	0.860	0.848
50%	36.12 dB	0.869	0.863

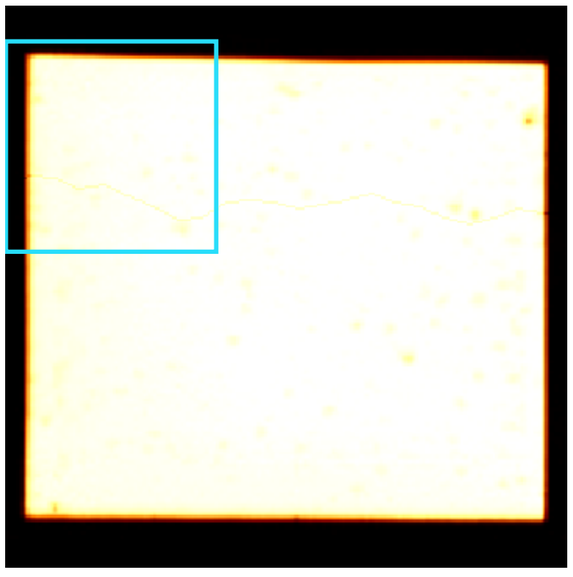
Crop: x=160:256, y=144:240. All panels use the same coordinates and intensity scale.



Organic / image0288 | Full 256x256 field

HELD-OUT TEST SPLIT

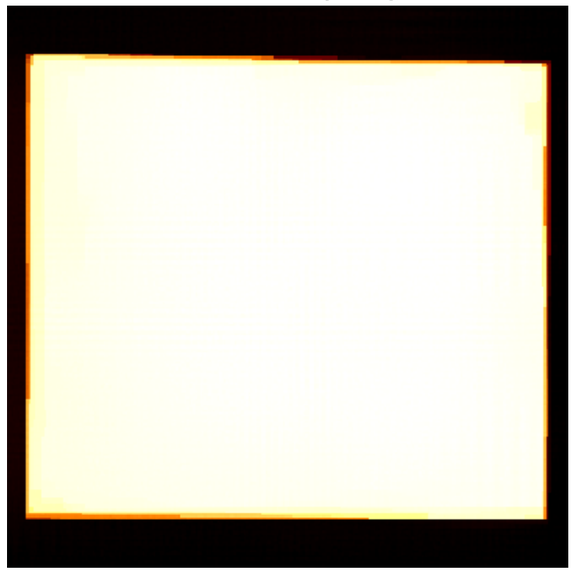
Ground truth



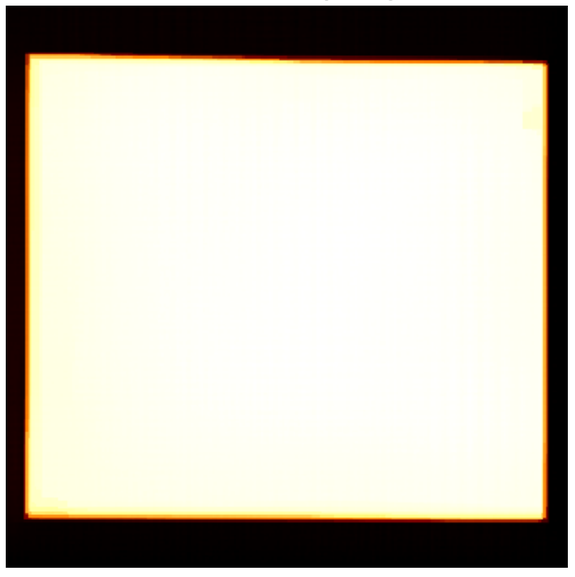
5% sampling



10% sampling



20% sampling



30% sampling



40% sampling



50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

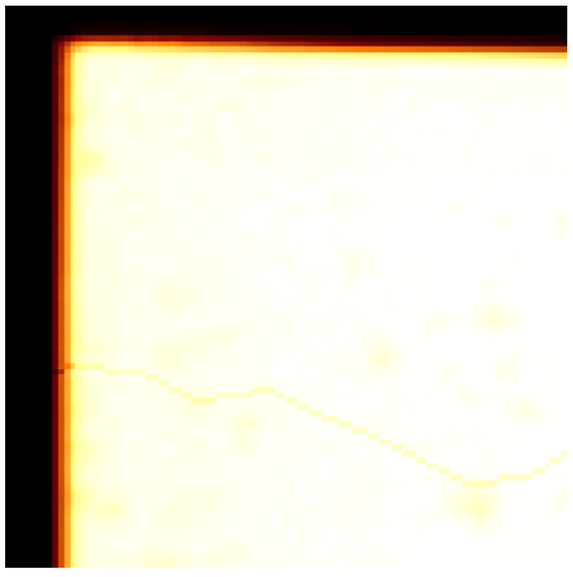
	PSNR	SSIM	HaarPSI
5%	26.17 dB	0.806	0.551
10%	29.52 dB	0.826	0.649
20%	33.40 dB	0.845	0.826
30%	34.55 dB	0.854	0.849
40%	33.29 dB	0.839	0.831
50%	35.19 dB	0.854	0.850

Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.

Organic / image0288 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

Ground truth



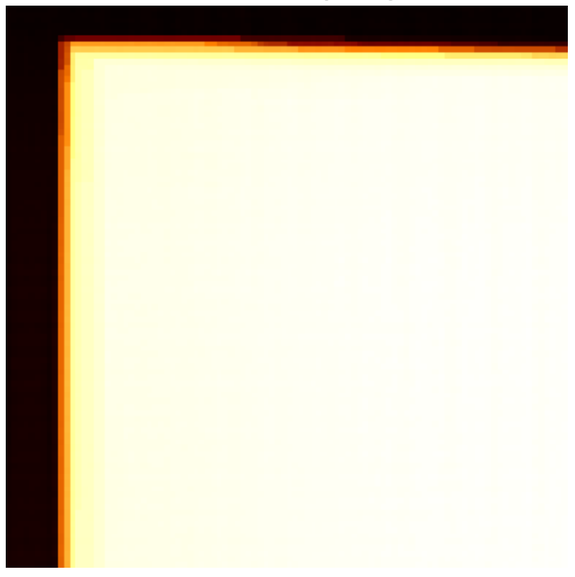
5% sampling



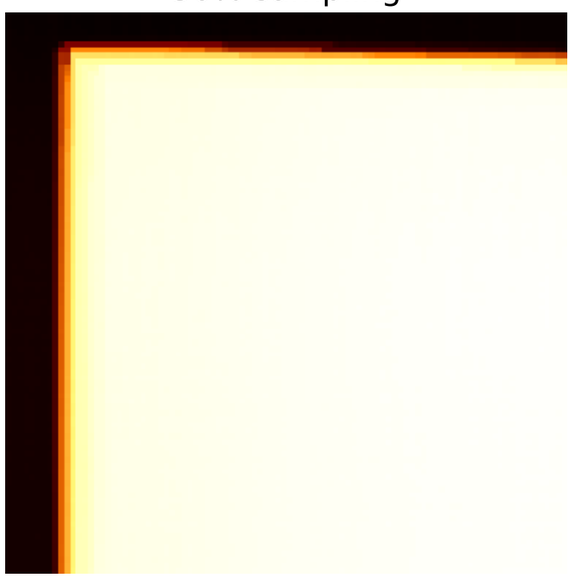
10% sampling



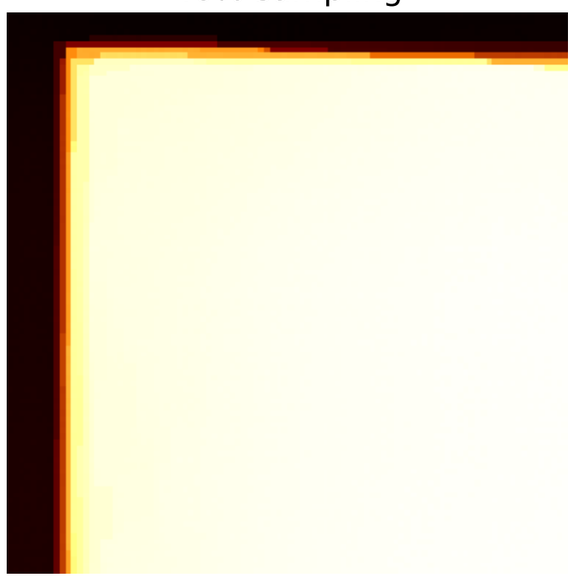
20% sampling



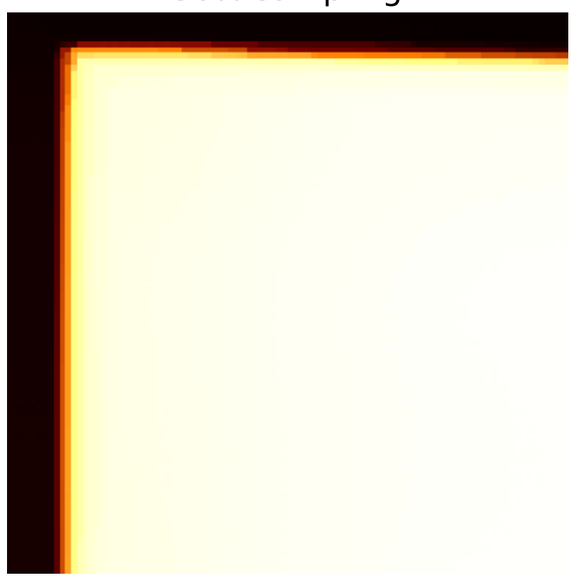
30% sampling



40% sampling



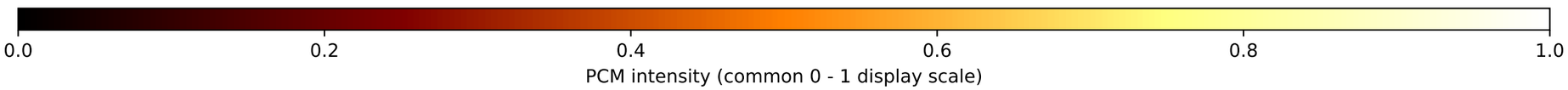
50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI			
5%	26.17 dB	0.806	0.551
10%	29.52 dB	0.826	0.649
20%	33.40 dB	0.845	0.826
30%	34.55 dB	0.854	0.849
40%	33.29 dB	0.839	0.831
50%	35.19 dB	0.854	0.850

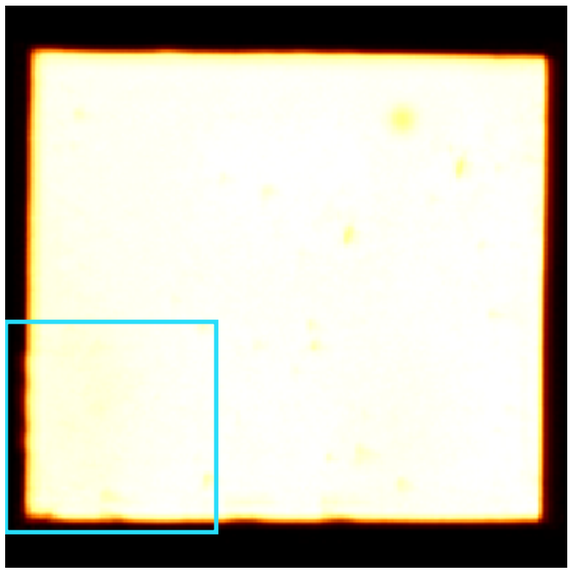
Crop: x=0:96, y=16:112. All panels use the same coordinates and intensity scale.



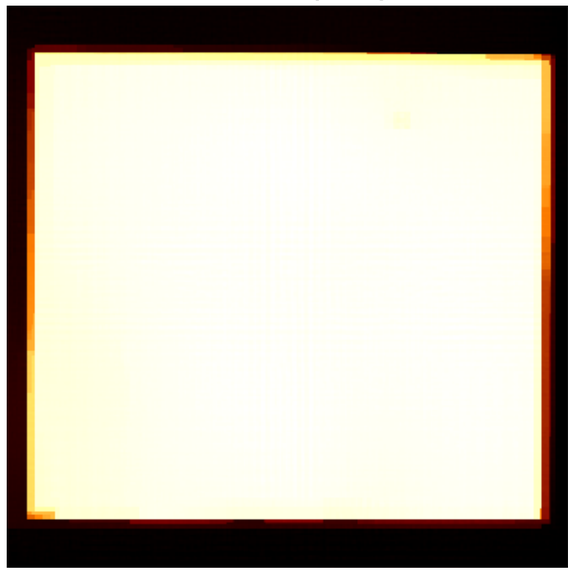
Organic / image0611 | Full 256x256 field

HELD-OUT TEST SPLIT

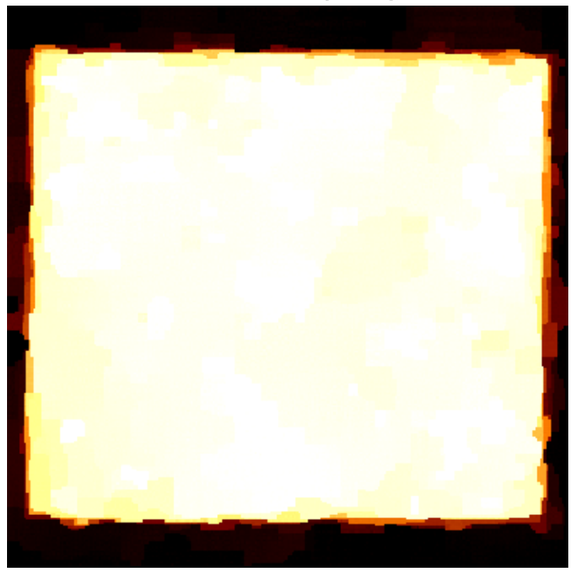
Ground truth



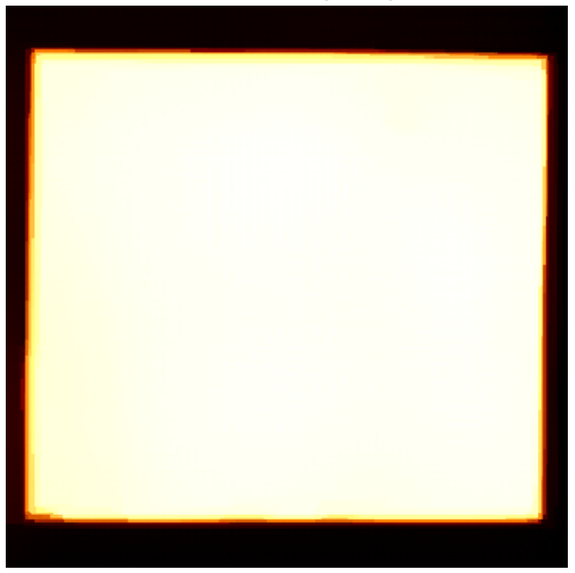
5% sampling



10% sampling



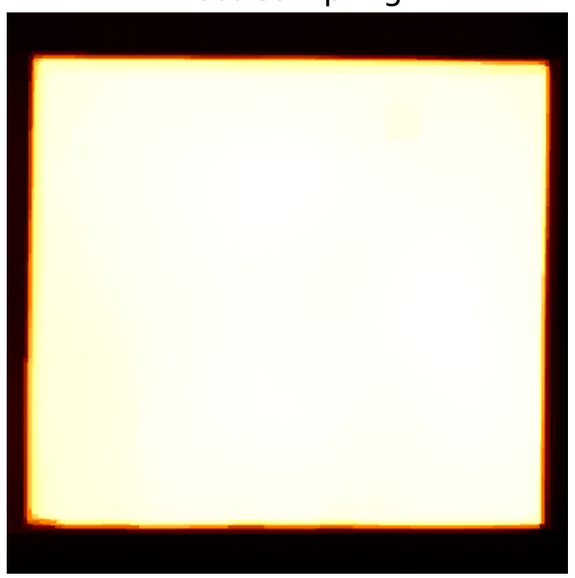
20% sampling



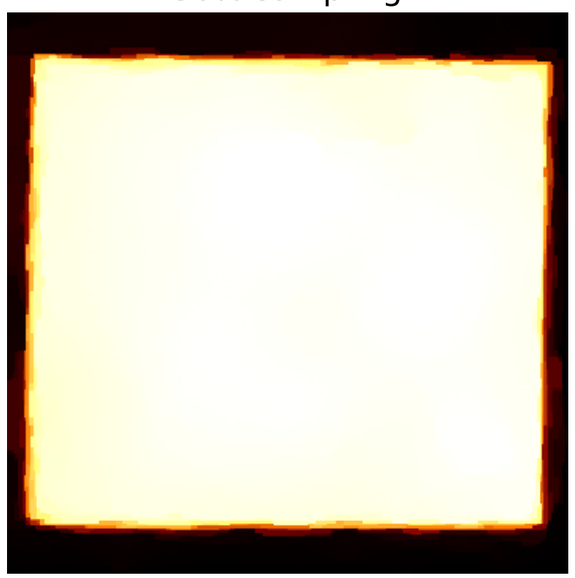
30% sampling



40% sampling



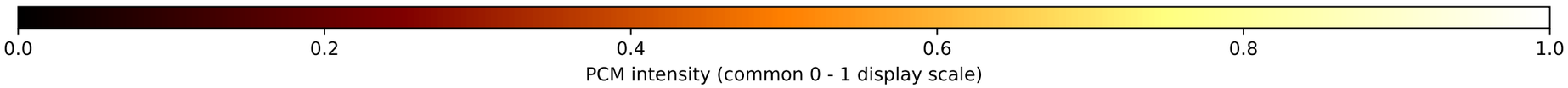
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   26.22 dB  0.833   0.501
10%   24.88 dB  0.738   0.461
20%   32.42 dB  0.867   0.772
30%   34.25 dB  0.883   0.825
40%   34.39 dB  0.885   0.807
50%   29.59 dB  0.846   0.687
```

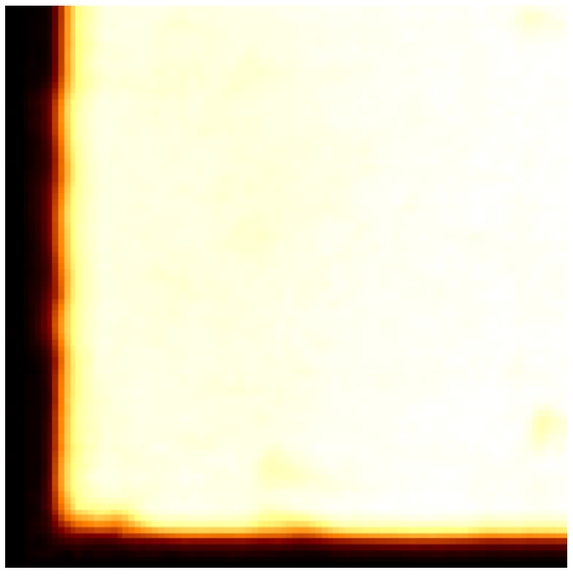
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



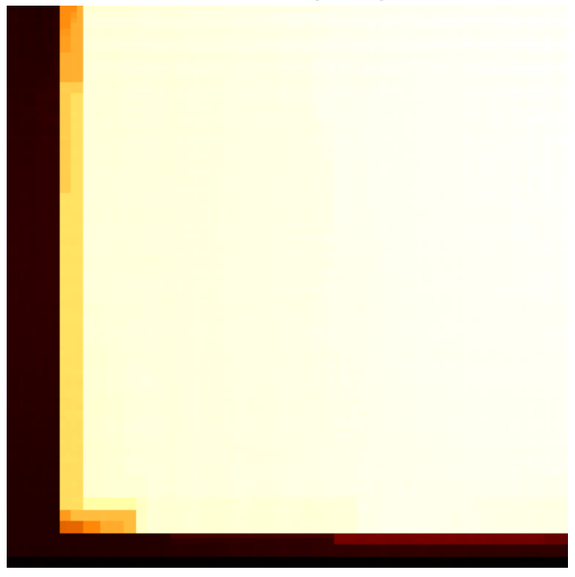
Organic / image0611 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

Ground truth



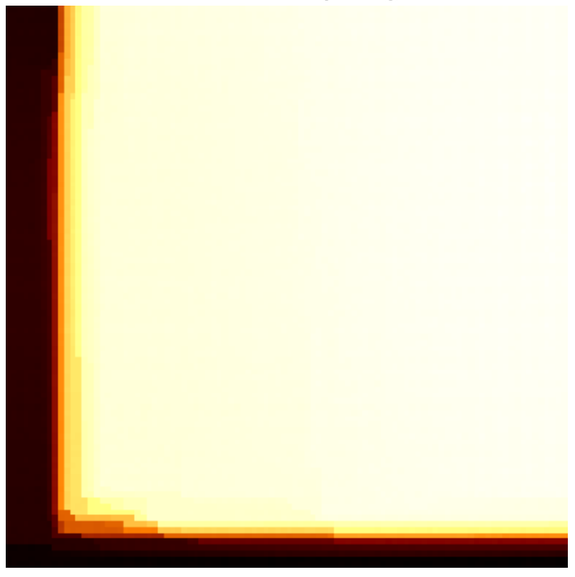
5% sampling



10% sampling



20% sampling



30% sampling



40% sampling



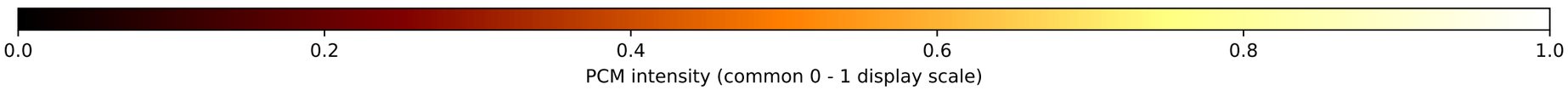
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   26.22 dB  0.833   0.501
10%   24.88 dB  0.738   0.461
20%   32.42 dB  0.867   0.772
30%   34.25 dB  0.883   0.825
40%   34.39 dB  0.885   0.807
50%   29.59 dB  0.846   0.687
```

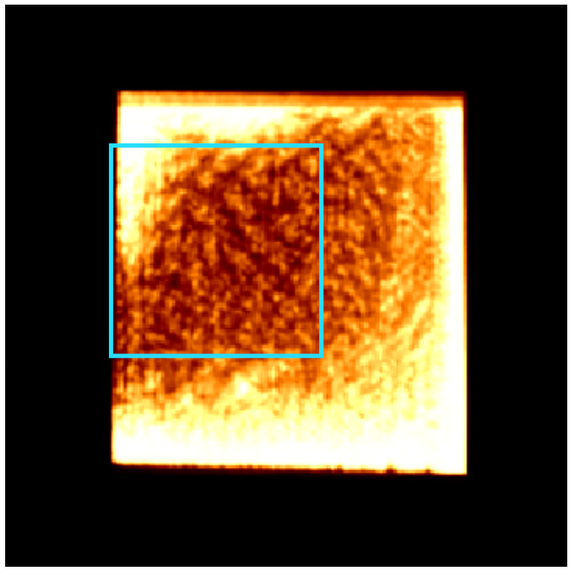
Crop: x=0:96, y=144:240. All panels use the same coordinates and intensity scale.



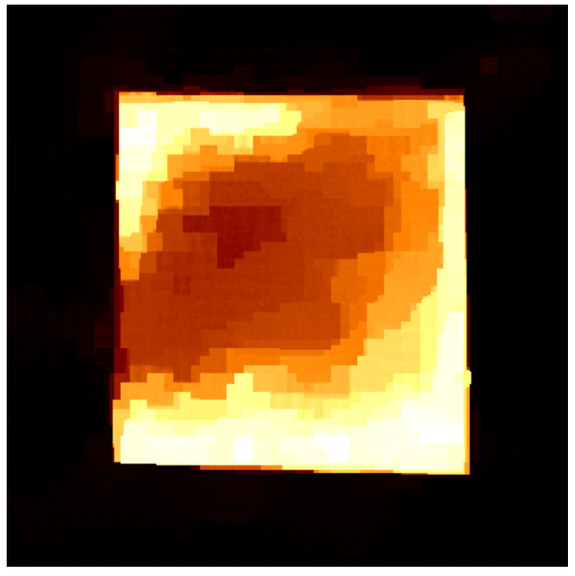
Perovskite / image0249 | Full 256x256 field

HELD-OUT TEST SPLIT

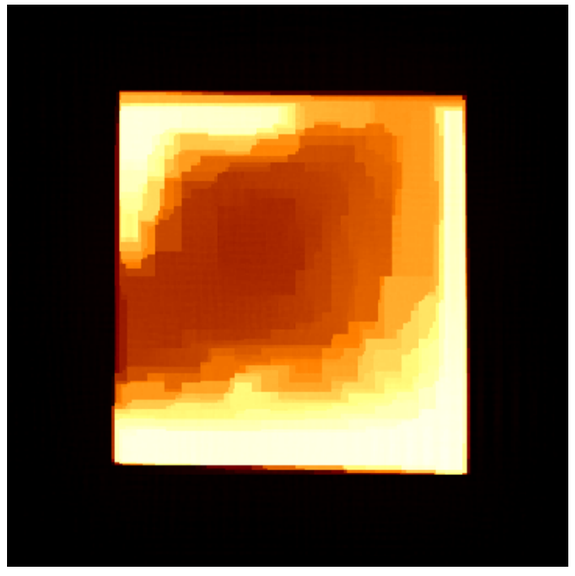
Ground truth



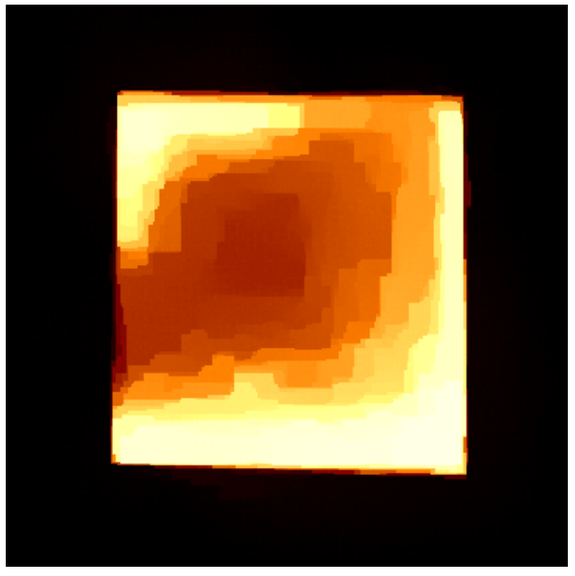
5% sampling



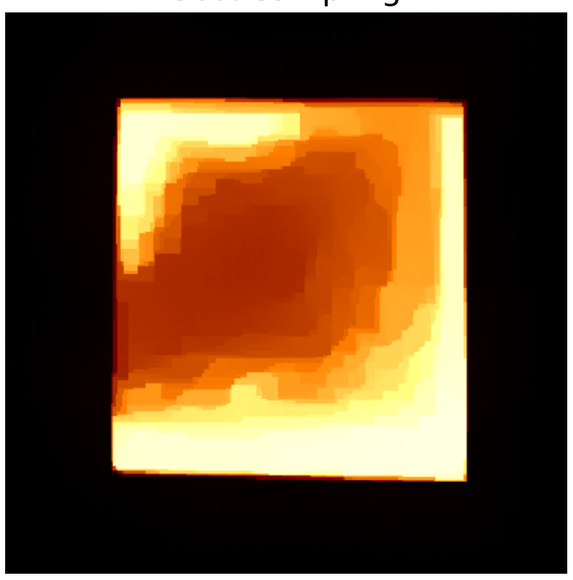
10% sampling



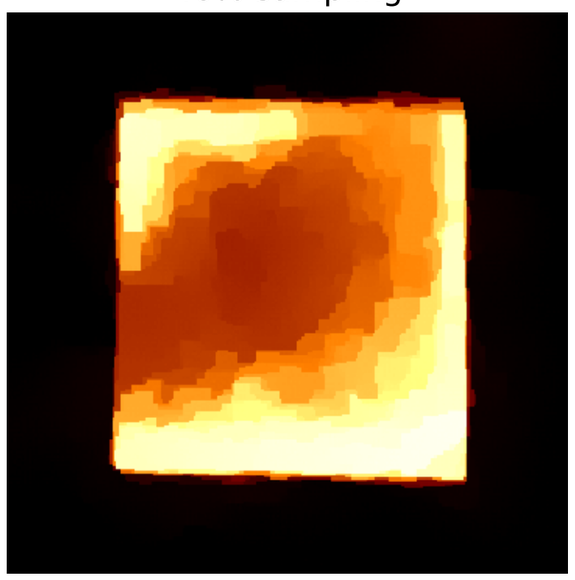
20% sampling



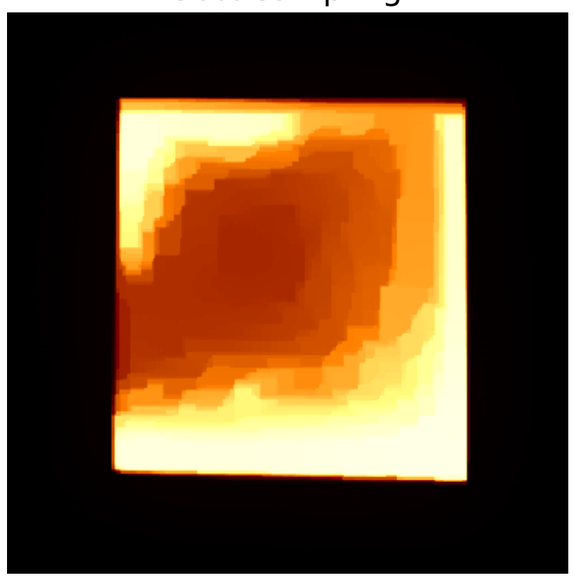
30% sampling



40% sampling



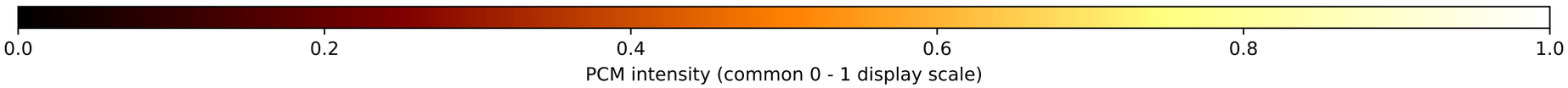
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   24.40 dB  0.535   0.373
10%   26.52 dB  0.645   0.450
20%   25.46 dB  0.612   0.415
30%   26.46 dB  0.620   0.439
40%   24.69 dB  0.554   0.387
50%   27.41 dB  0.668   0.493
```

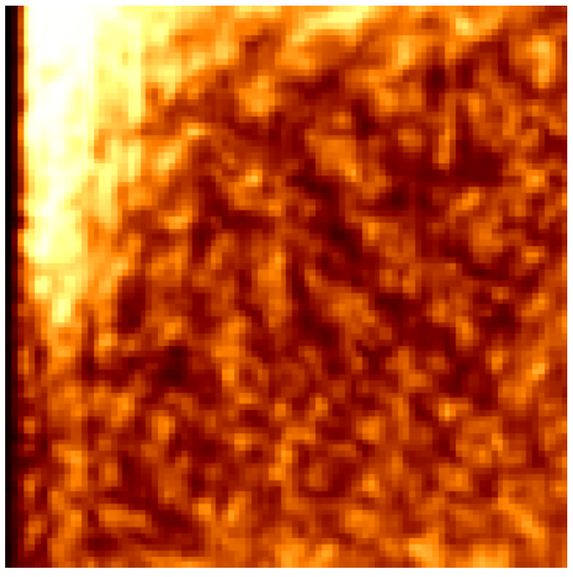
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



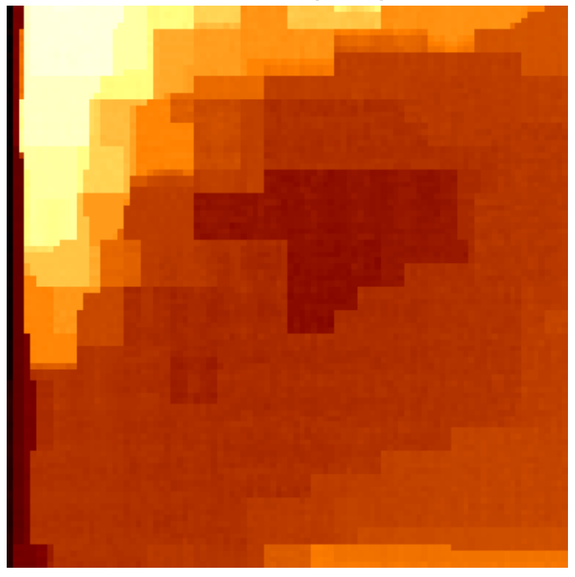
Perovskite / image0249 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

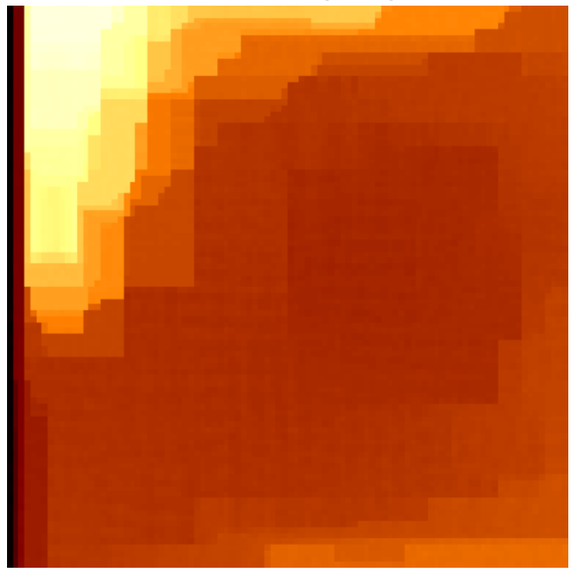
Ground truth



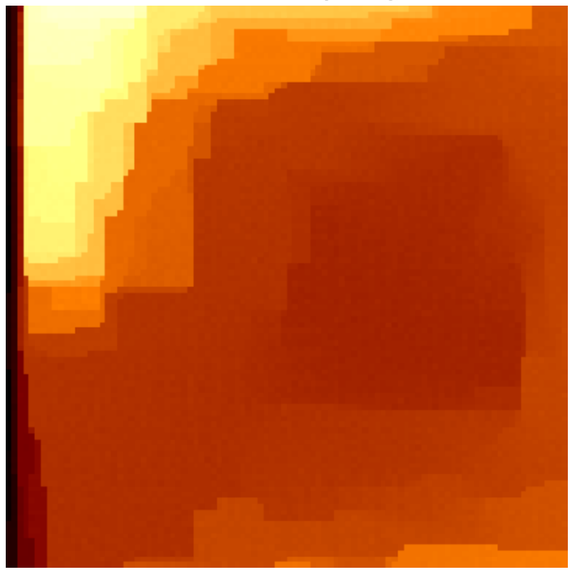
5% sampling



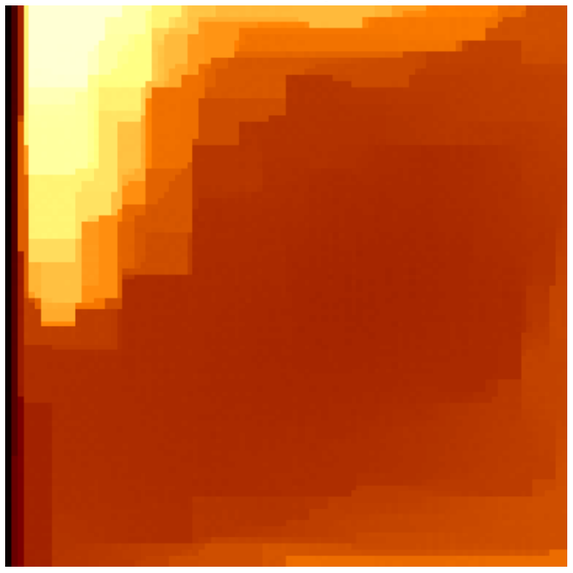
10% sampling



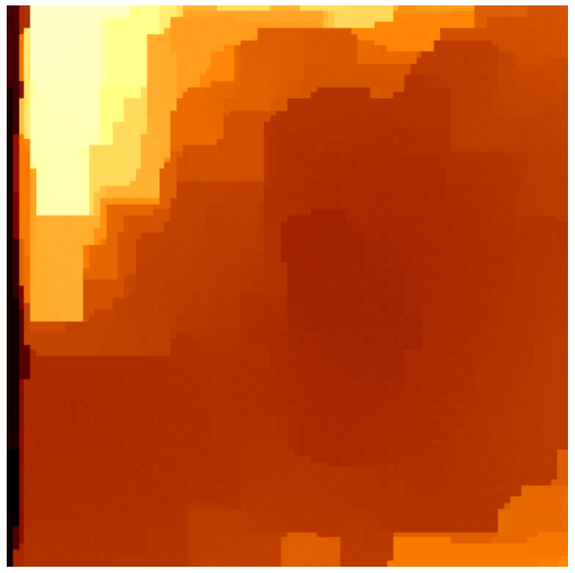
20% sampling



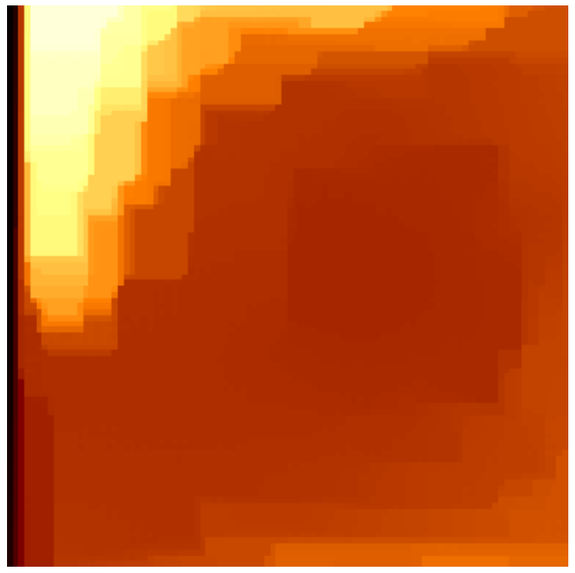
30% sampling



40% sampling



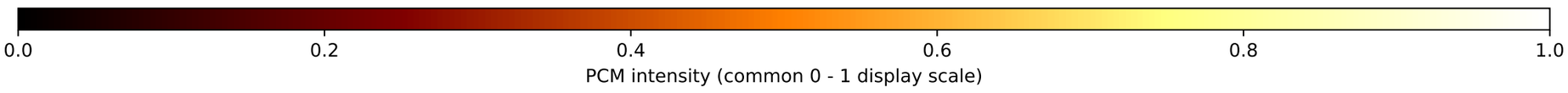
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   24.40 dB  0.535   0.373
10%   26.52 dB  0.645   0.450
20%   25.46 dB  0.612   0.415
30%   26.46 dB  0.620   0.439
40%   24.69 dB  0.554   0.387
50%   27.41 dB  0.668   0.493
```

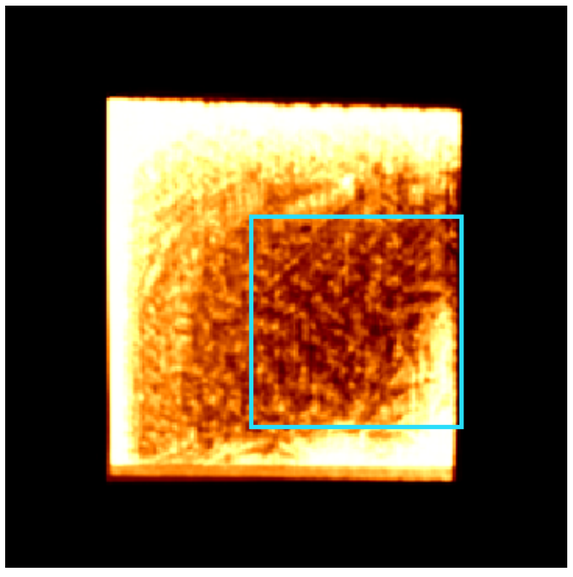
Crop: x=48:144, y=64:160. All panels use the same coordinates and intensity scale.



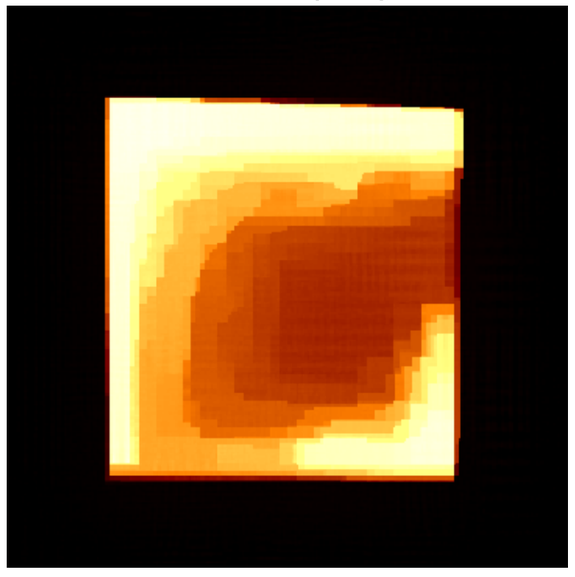
Perovskite / image0288 | Full 256x256 field

HELD-OUT TEST SPLIT

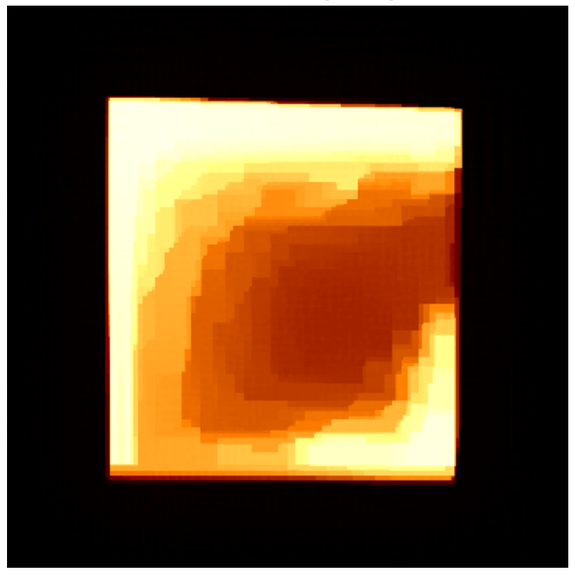
Ground truth



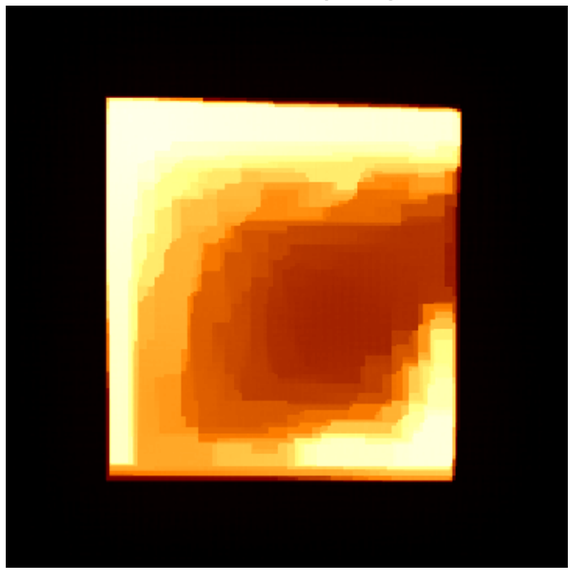
5% sampling



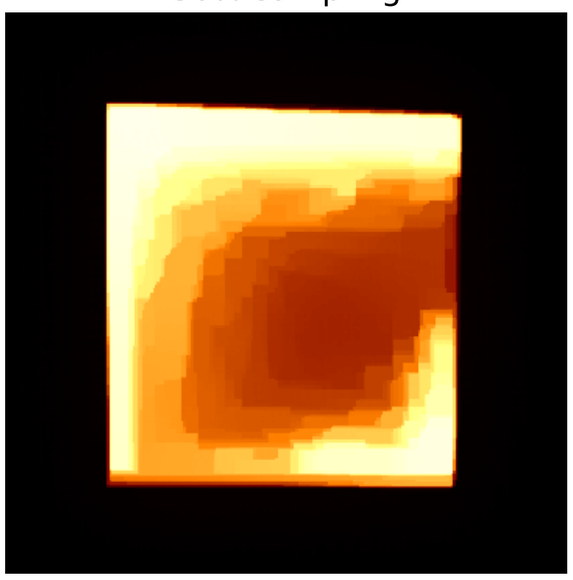
10% sampling



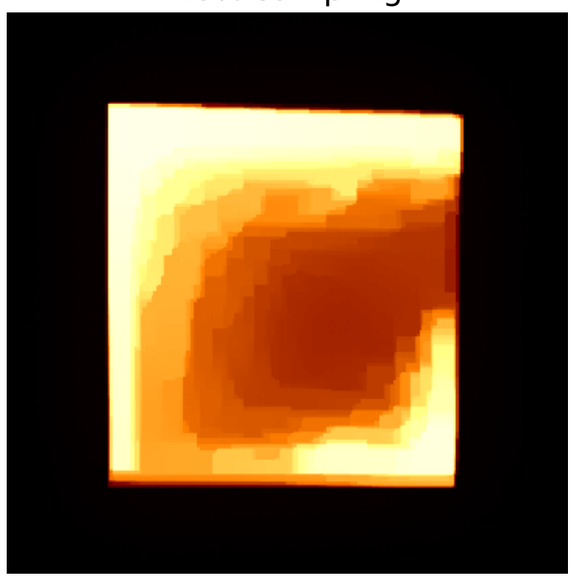
20% sampling



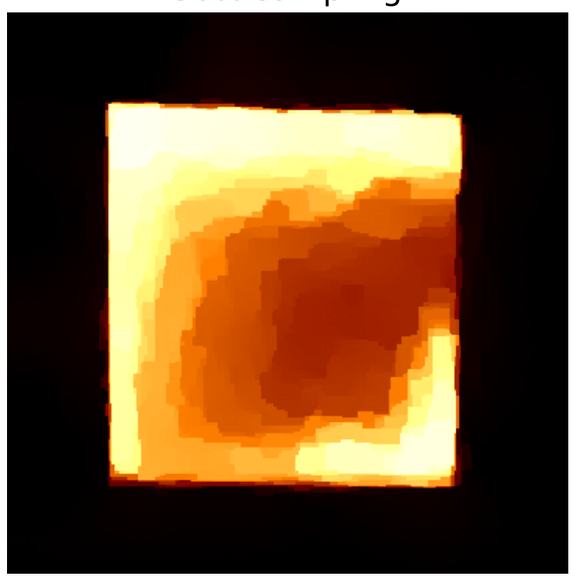
30% sampling



40% sampling



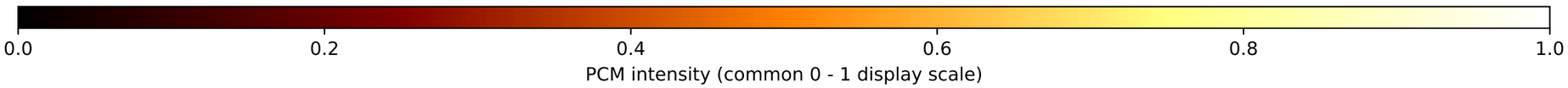
50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

	PSNR	SSIM	HaarPSI
5%	25.30 dB	0.629	0.431
10%	26.59 dB	0.652	0.465
20%	27.18 dB	0.655	0.477
30%	27.43 dB	0.678	0.488
40%	27.44 dB	0.666	0.496
50%	25.01 dB	0.547	0.402

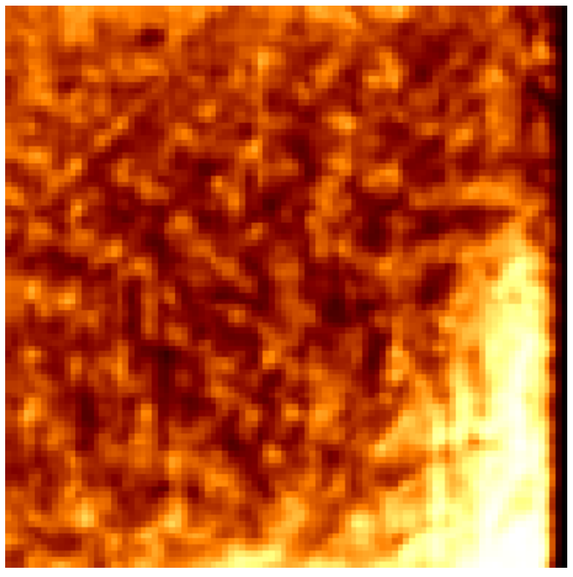
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



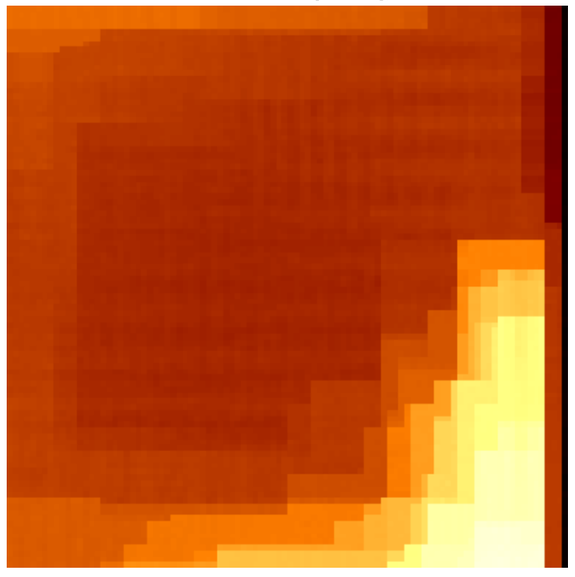
Perovskite / image0288 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

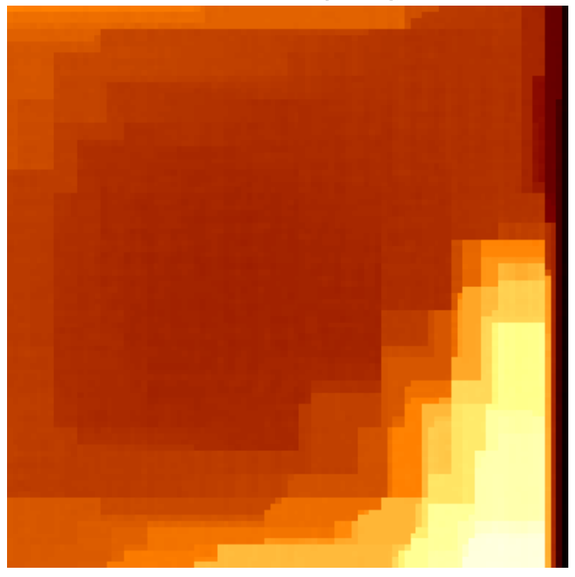
Ground truth



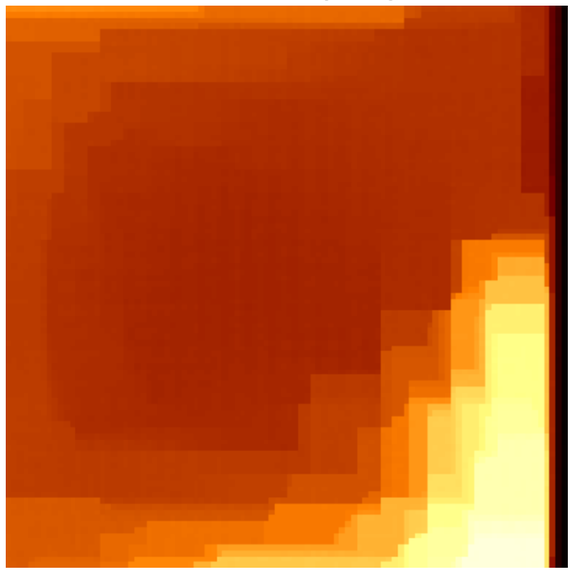
5% sampling



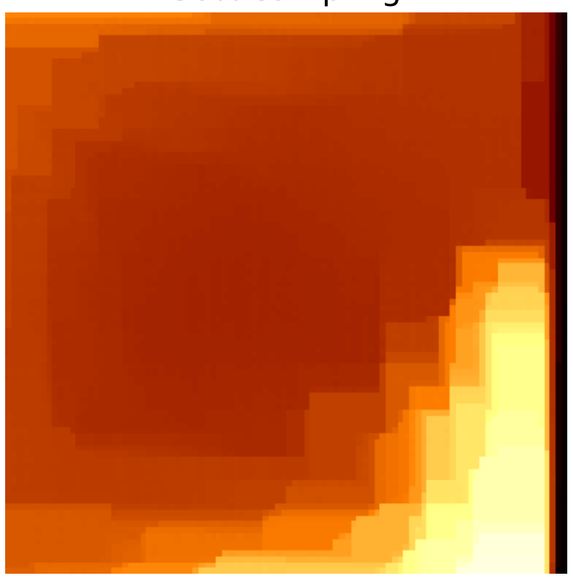
10% sampling



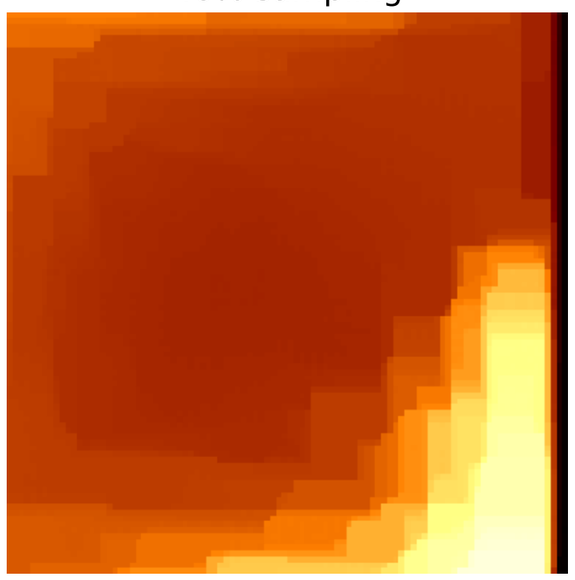
20% sampling



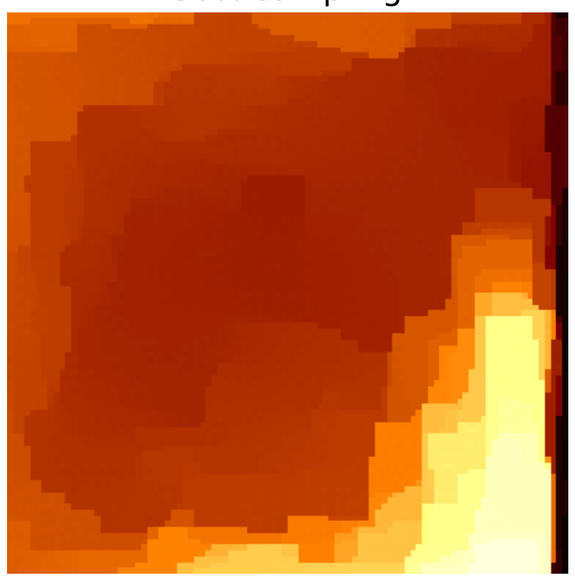
30% sampling



40% sampling



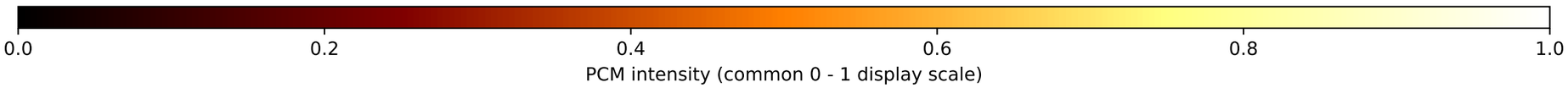
50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI			
5%	25.30 dB	0.629	0.431
10%	26.59 dB	0.652	0.465
20%	27.18 dB	0.655	0.477
30%	27.43 dB	0.678	0.488
40%	27.44 dB	0.666	0.496
50%	25.01 dB	0.547	0.402

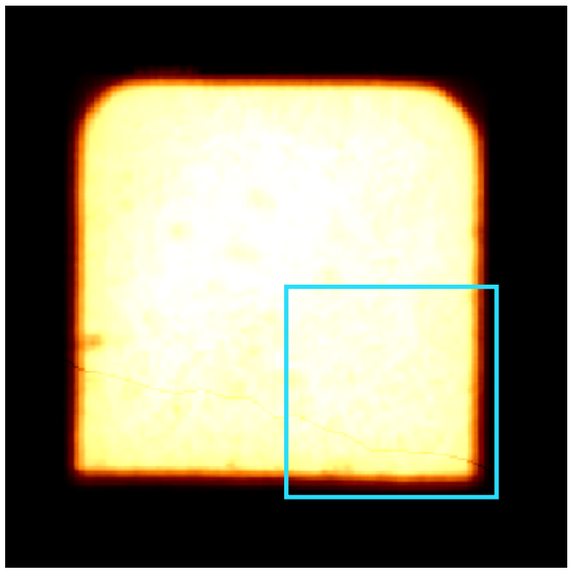
Crop: x=112:208, y=96:192. All panels use the same coordinates and intensity scale.



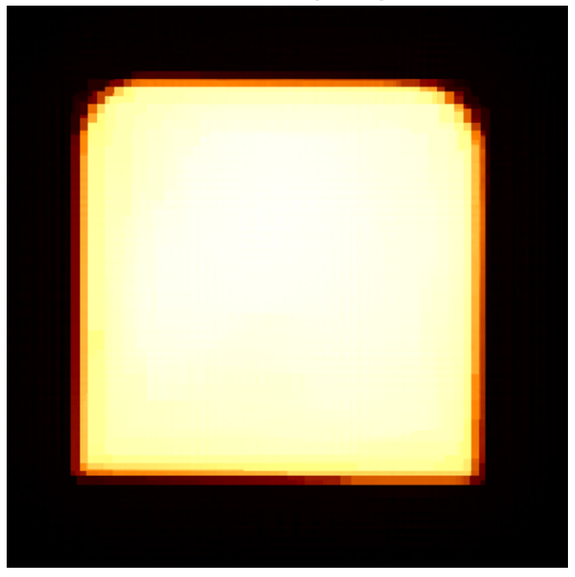
Perovskite / image0611 | Full 256x256 field

HELD-OUT TEST SPLIT

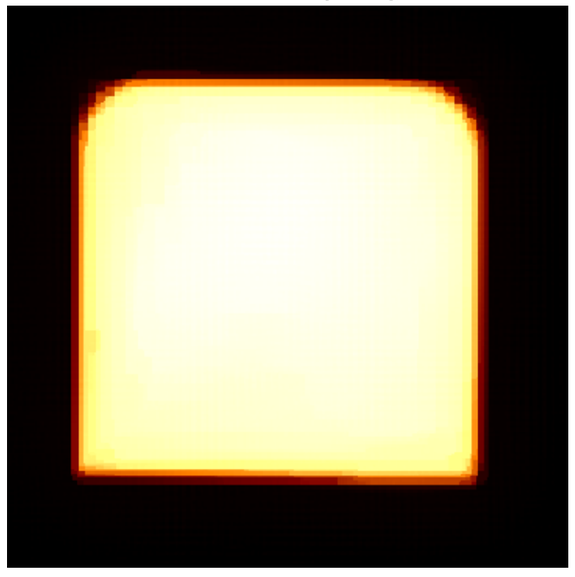
Ground truth



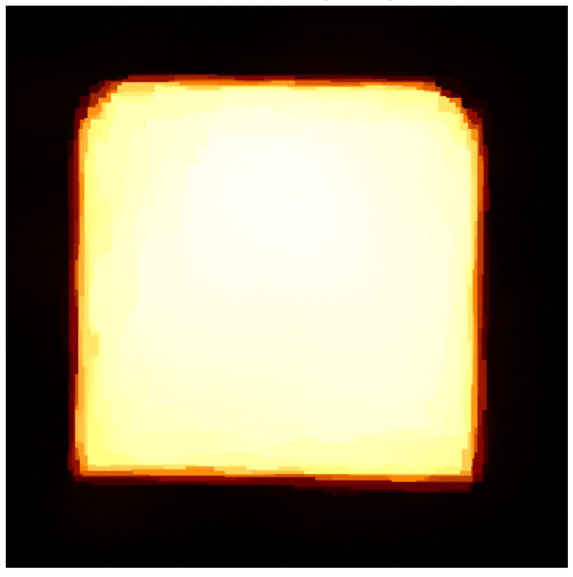
5% sampling



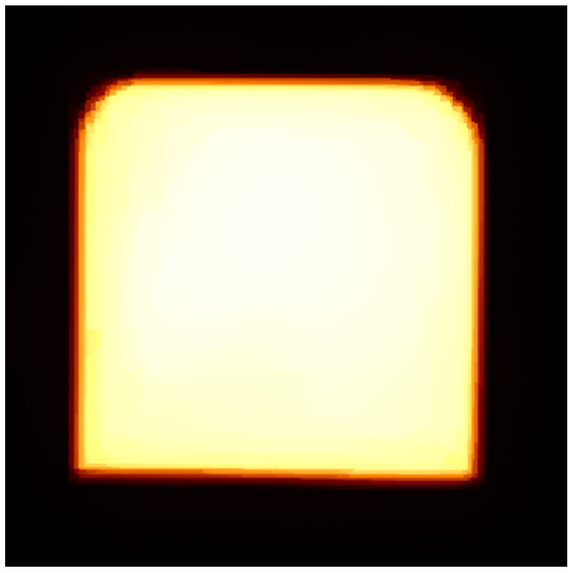
10% sampling



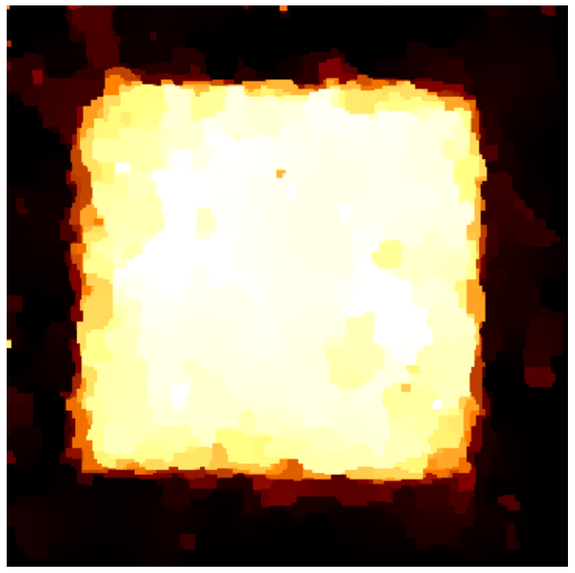
20% sampling



30% sampling



40% sampling



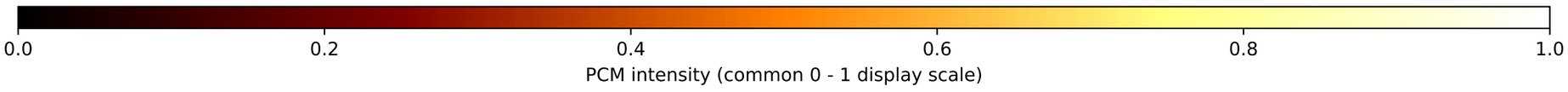
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   30.02 dB  0.722   0.614
10%   31.27 dB  0.735   0.716
20%   30.76 dB  0.697   0.684
30%   34.27 dB  0.754   0.797
40%   22.25 dB  0.482   0.328
50%   35.05 dB  0.759   0.821
```

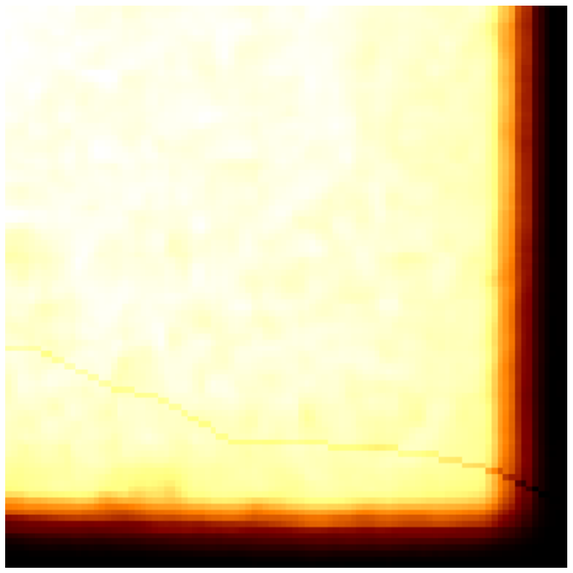
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



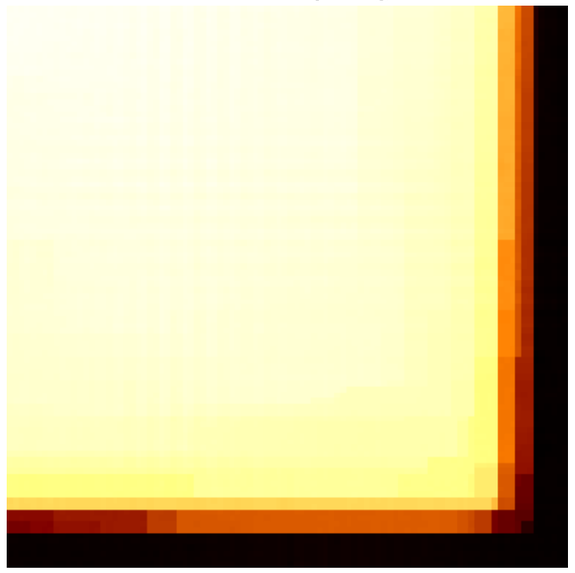
Perovskite / image0611 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

Ground truth



5% sampling



10% sampling



20% sampling



30% sampling



40% sampling



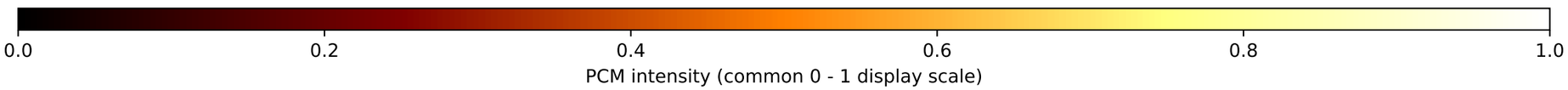
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   30.02 dB  0.722   0.614
10%   31.27 dB  0.735   0.716
20%   30.76 dB  0.697   0.684
30%   34.27 dB  0.754   0.797
40%   22.25 dB  0.482   0.328
50%   35.05 dB  0.759   0.821
```

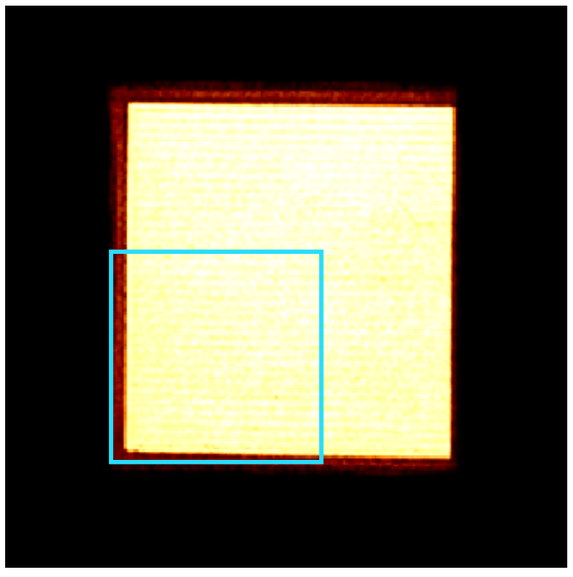
Crop: x=128:224, y=128:224. All panels use the same coordinates and intensity scale.



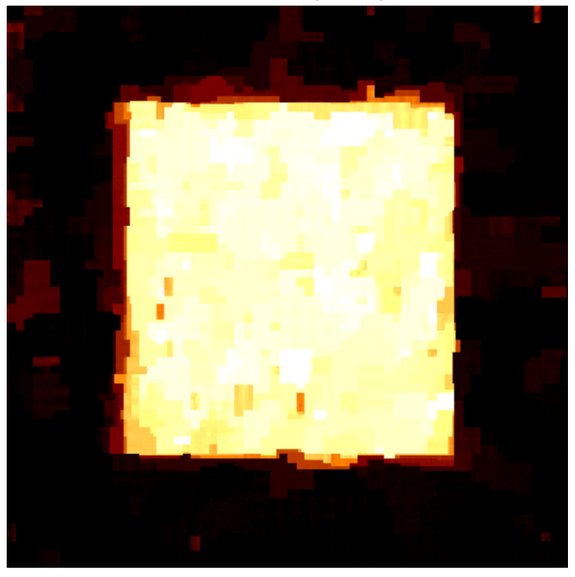
Reference / image0249 | Full 256x256 field

HELD-OUT TEST SPLIT

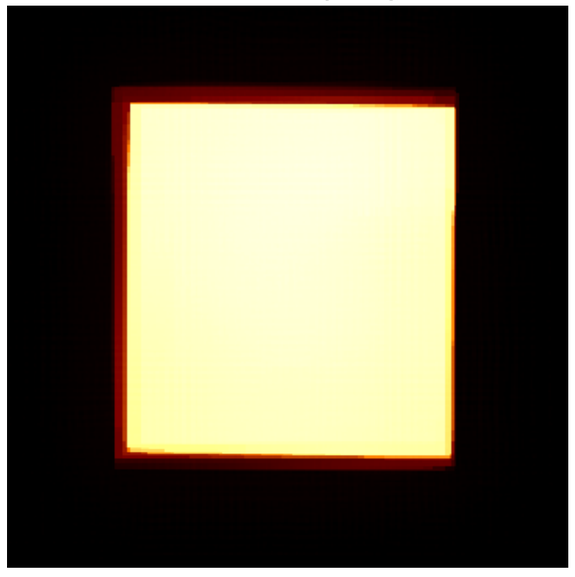
Ground truth



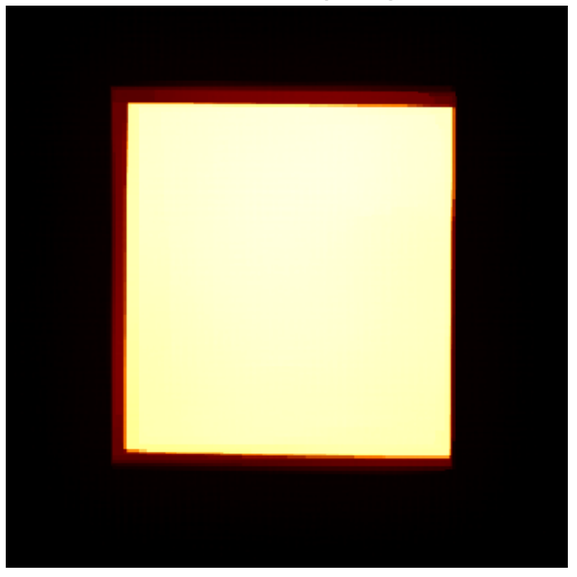
5% sampling



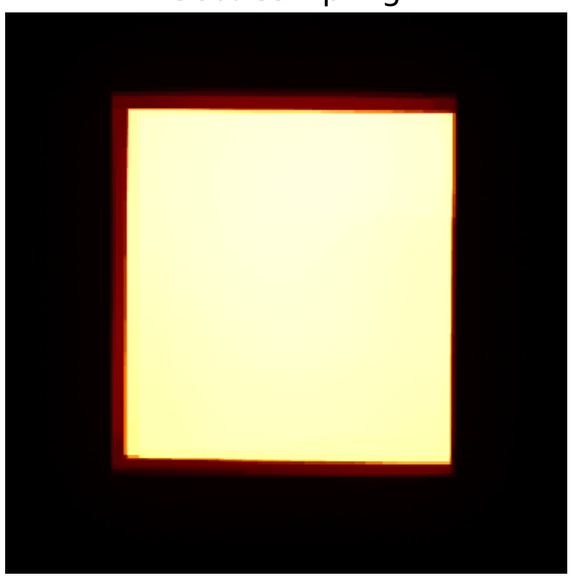
10% sampling



20% sampling



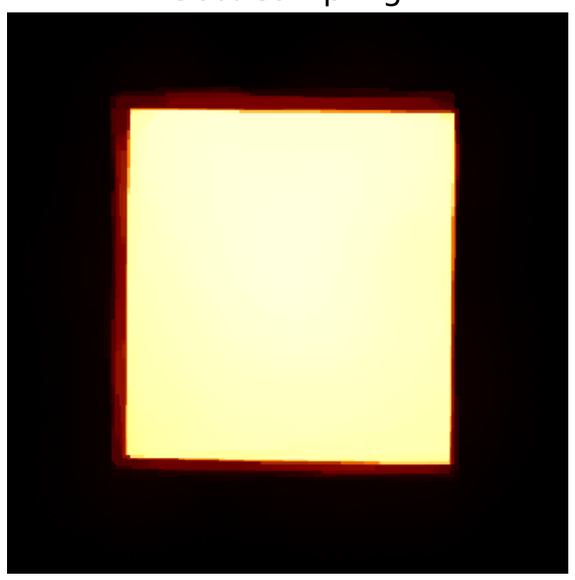
30% sampling



40% sampling



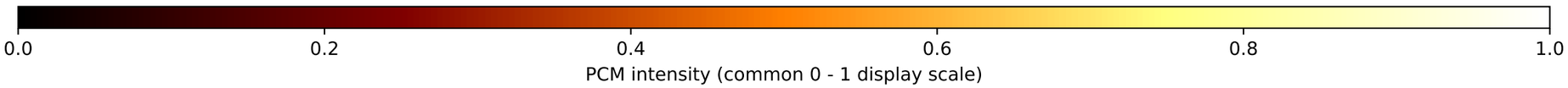
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   22.24 dB  0.343   0.331
10%   29.77 dB  0.648   0.586
20%   31.40 dB  0.658   0.639
30%   31.57 dB  0.658   0.636
40%   23.41 dB  0.414   0.343
50%   30.84 dB  0.620   0.628
```

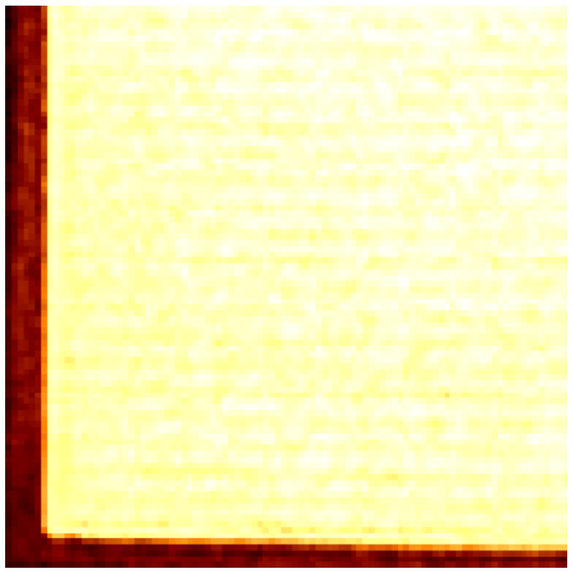
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



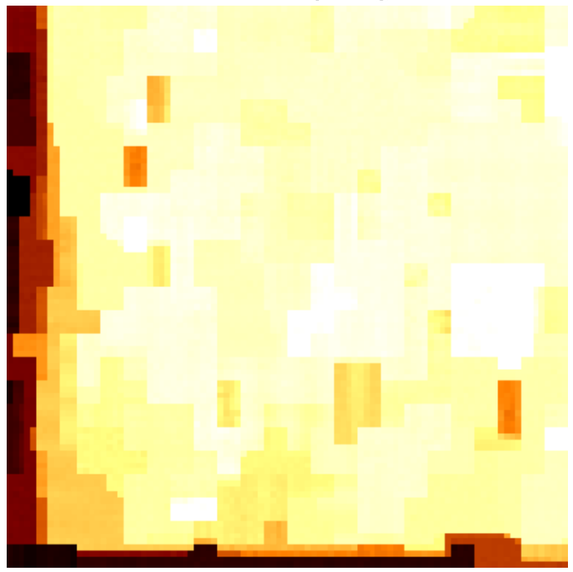
Reference / image0249 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

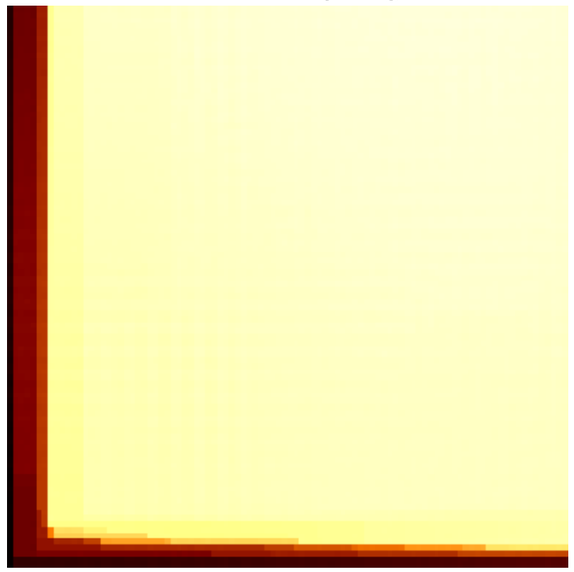
Ground truth



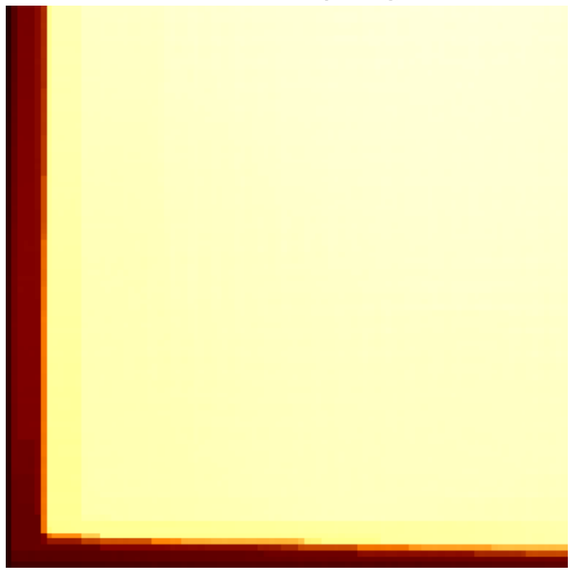
5% sampling



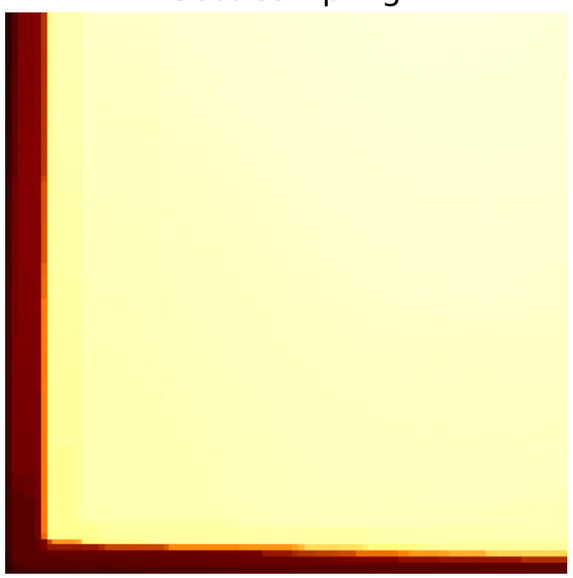
10% sampling



20% sampling



30% sampling



40% sampling



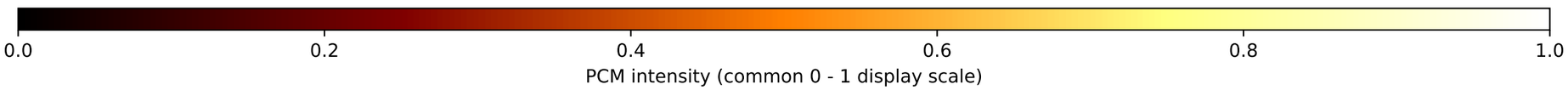
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   22.24 dB  0.343   0.331
10%   29.77 dB  0.648   0.586
20%   31.40 dB  0.658   0.639
30%   31.57 dB  0.658   0.636
40%   23.41 dB  0.414   0.343
50%   30.84 dB  0.620   0.628
```

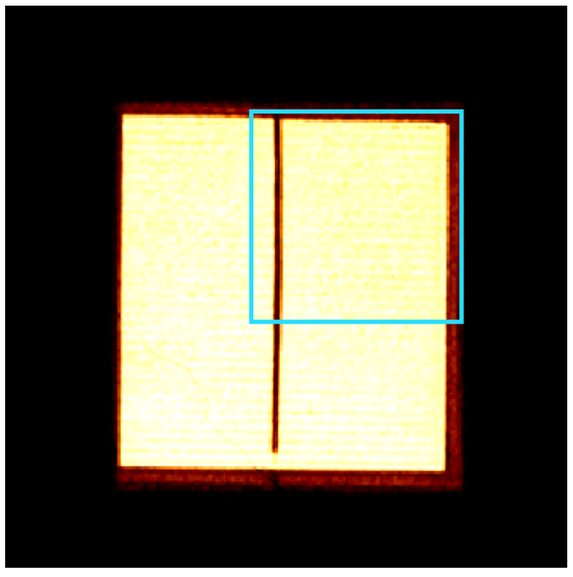
Crop: x=48:144, y=112:208. All panels use the same coordinates and intensity scale.



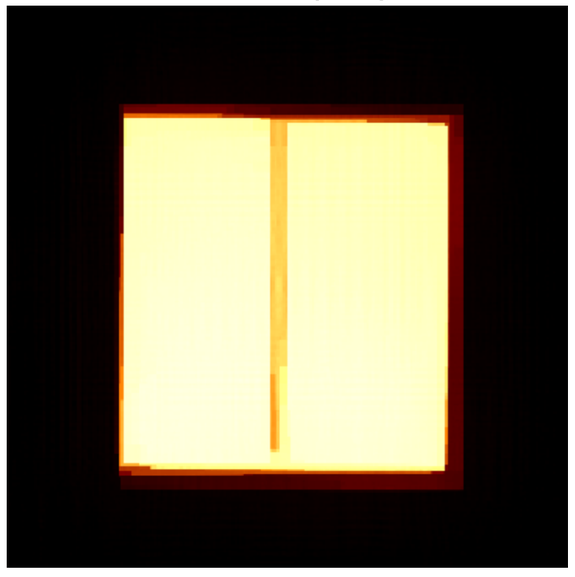
Reference / image0288 | Full 256x256 field

HELD-OUT TEST SPLIT

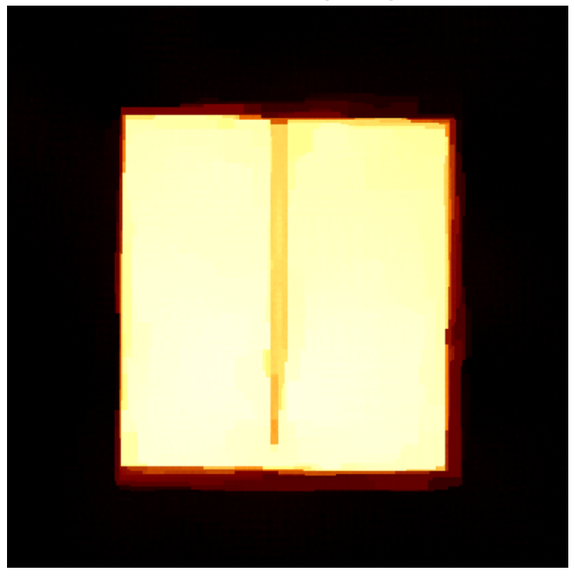
Ground truth



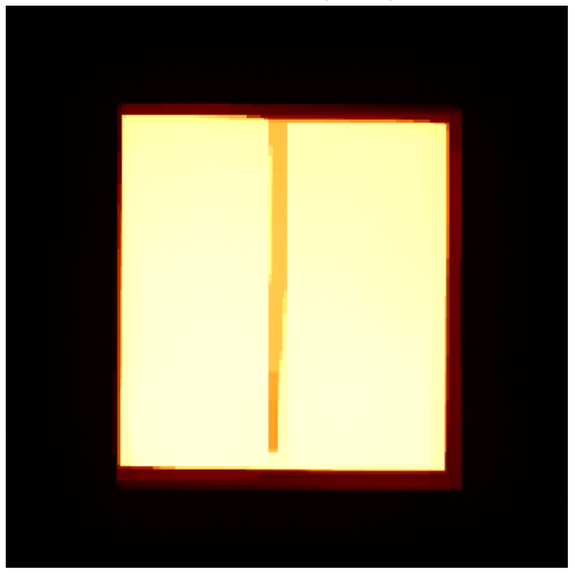
5% sampling



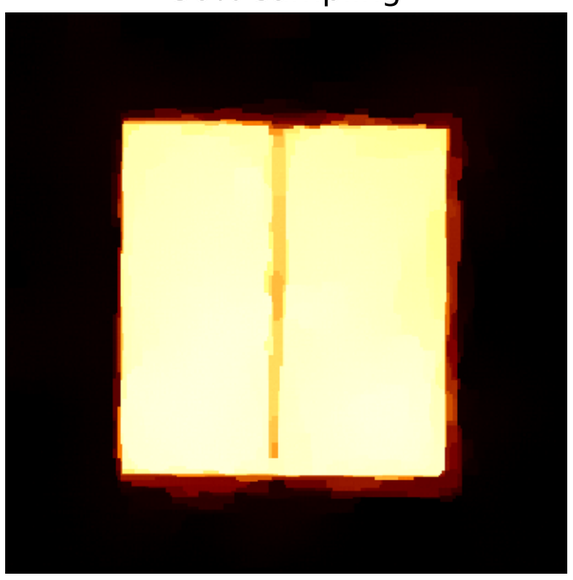
10% sampling



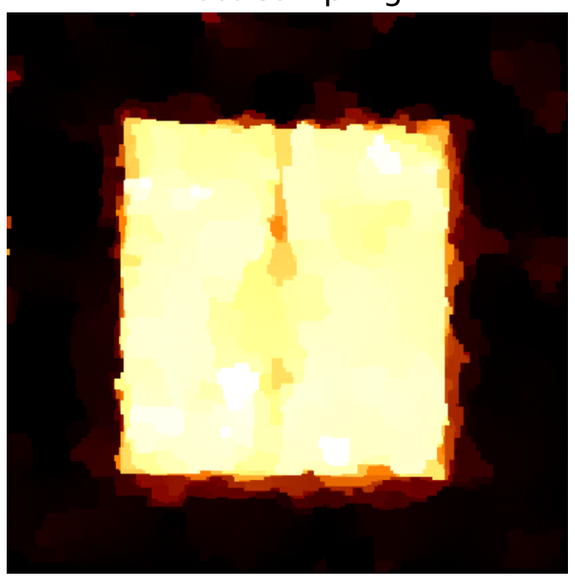
20% sampling



30% sampling



40% sampling



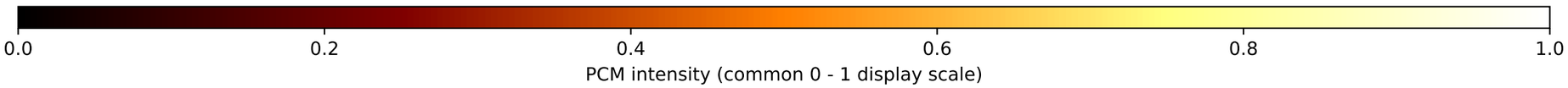
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

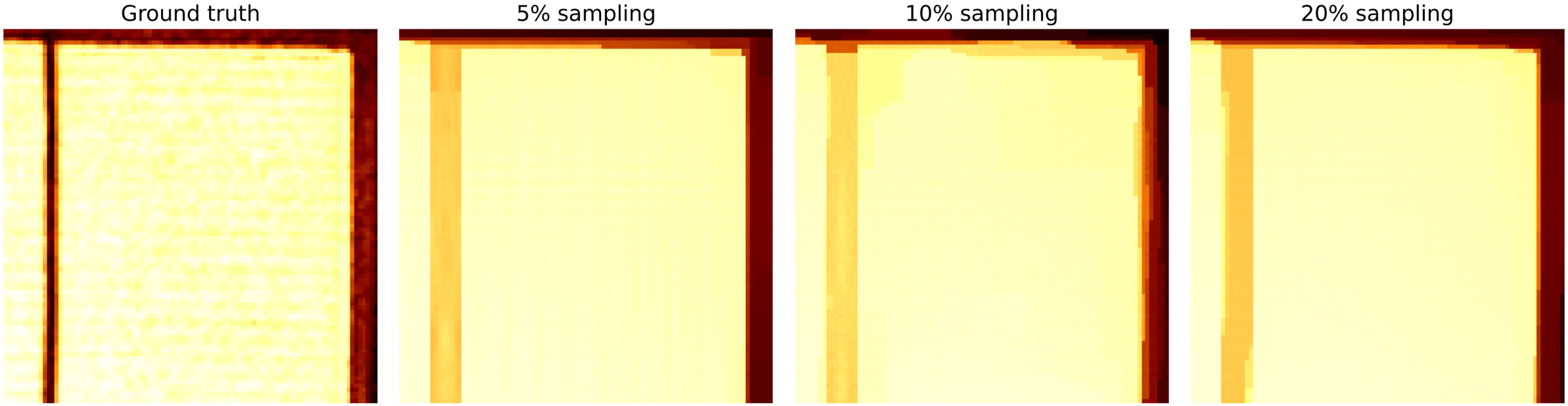
PSNR / SSIM / HaarPSI
 5%   25.24 dB  0.627   0.457
10%   25.03 dB  0.609   0.447
20%   26.54 dB  0.637   0.542
30%   25.42 dB  0.552   0.490
40%   21.61 dB  0.365   0.316
50%   24.76 dB  0.542   0.470
```

Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



Reference / image0288 | Local detail (automatically selected gradient-rich 96px crop)

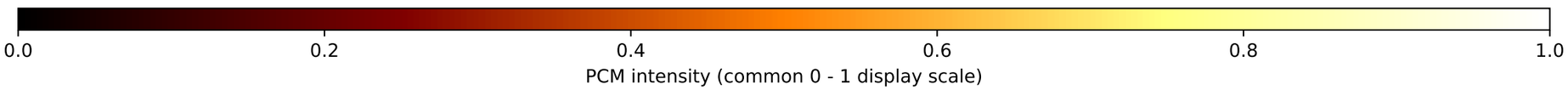
HELD-OUT TEST SPLIT



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

	PSNR	SSIM	HaarPSI
5%	25.24 dB	0.627	0.457
10%	25.03 dB	0.609	0.447
20%	26.54 dB	0.637	0.542
30%	25.42 dB	0.552	0.490
40%	21.61 dB	0.365	0.316
50%	24.76 dB	0.542	0.470

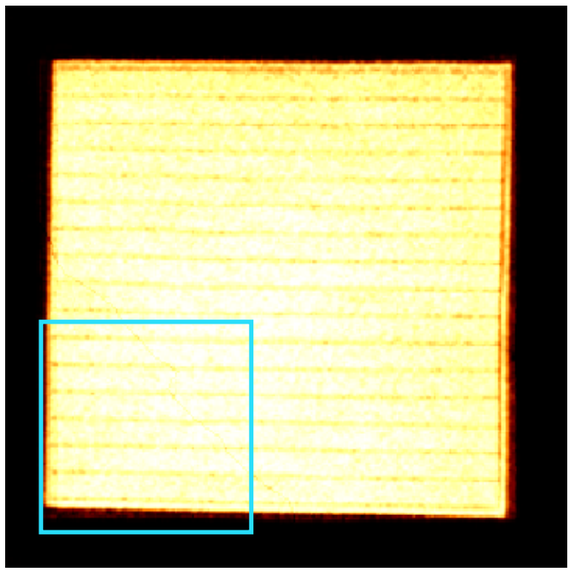
Crop: x=112:208, y=48:144. All panels use the same coordinates and intensity scale.



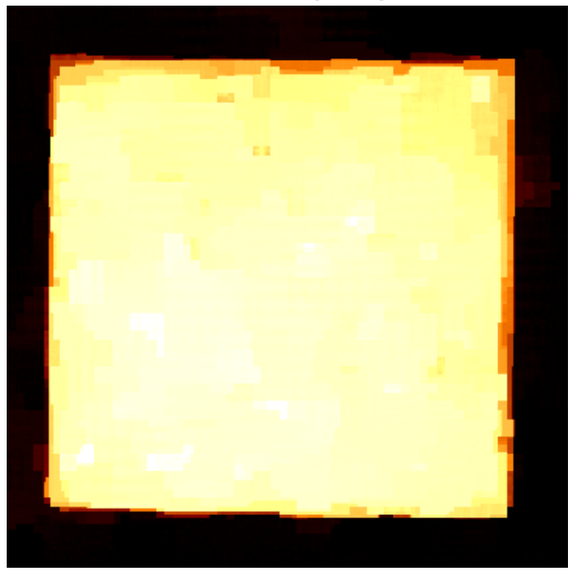
Reference / image0611 | Full 256x256 field

HELD-OUT TEST SPLIT

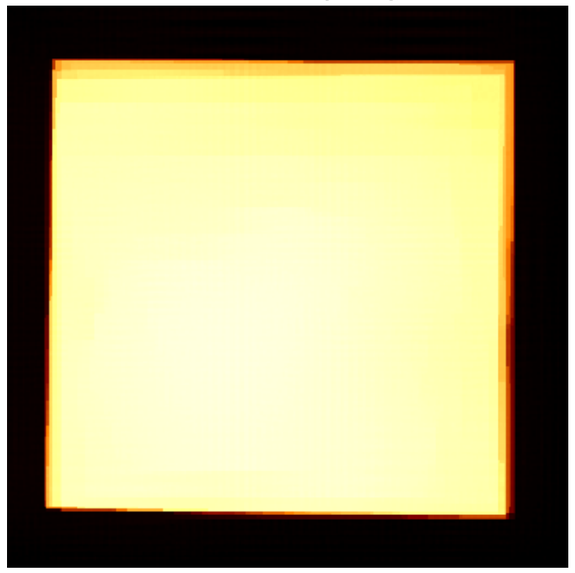
Ground truth



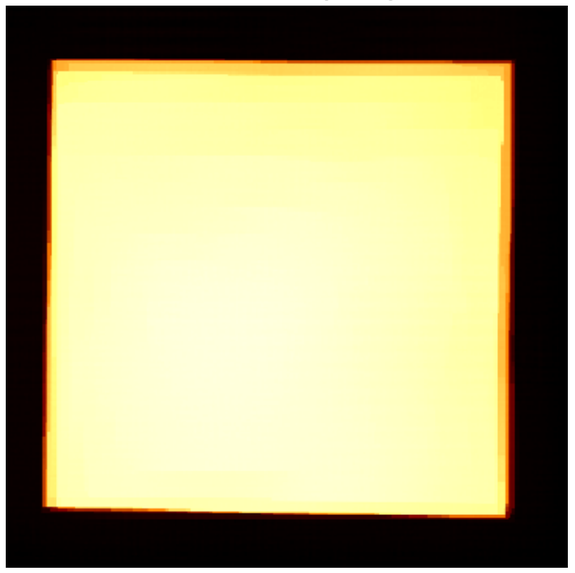
5% sampling



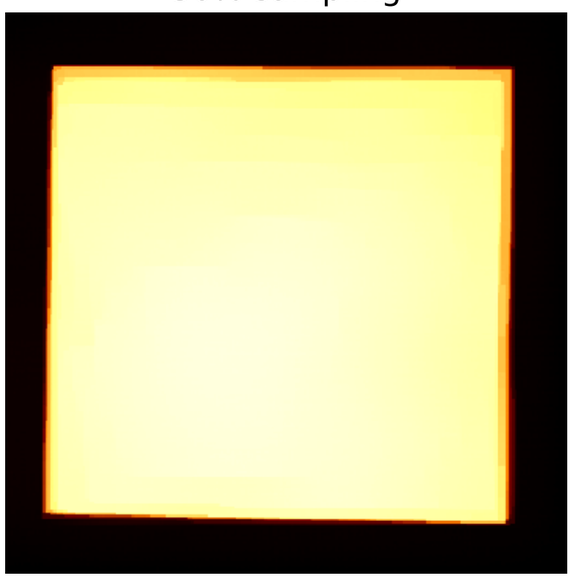
10% sampling



20% sampling



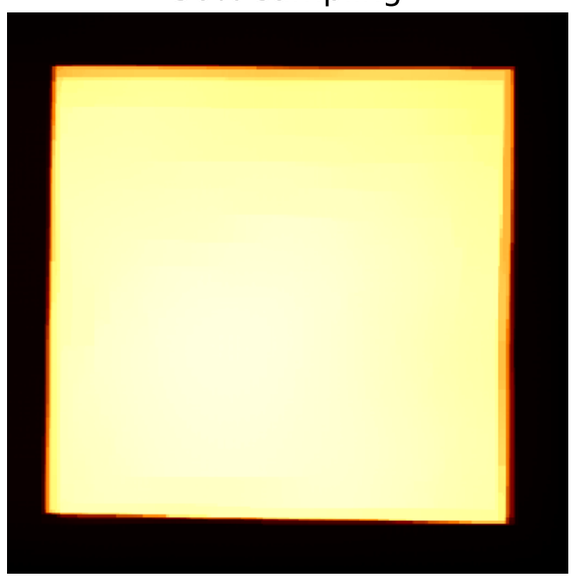
30% sampling



40% sampling



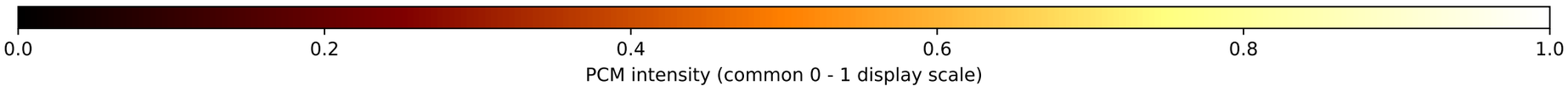
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   23.46 dB  0.411   0.365
10%   25.82 dB  0.448   0.396
20%   26.33 dB  0.452   0.409
30%   26.46 dB  0.449   0.398
40%   24.93 dB  0.428   0.359
50%   26.57 dB  0.459   0.402
```

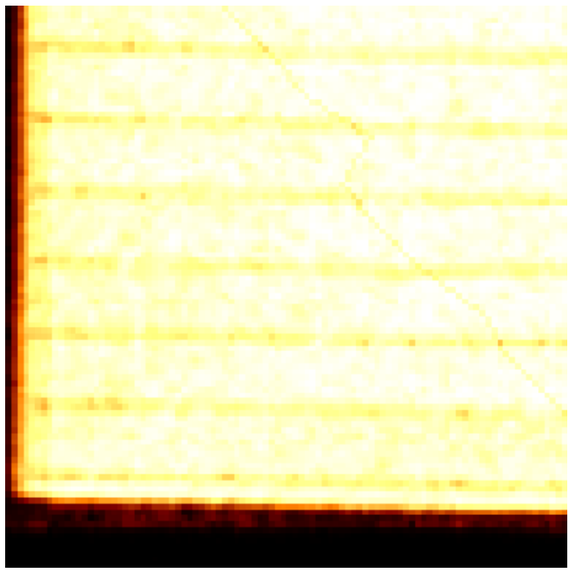
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



Reference / image0611 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

Ground truth



5% sampling



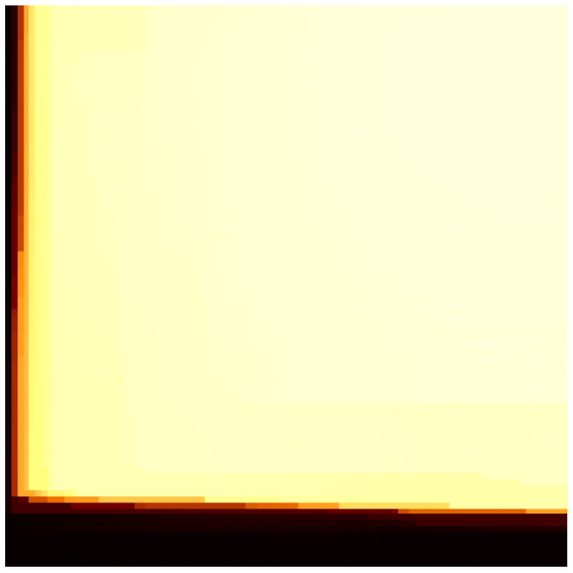
10% sampling



20% sampling



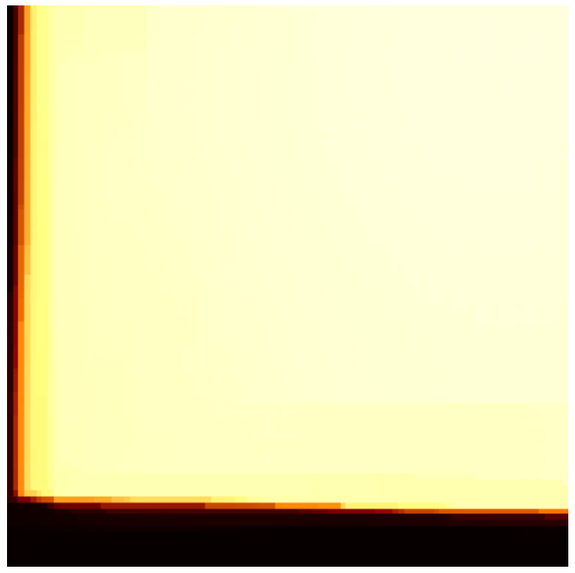
30% sampling



40% sampling



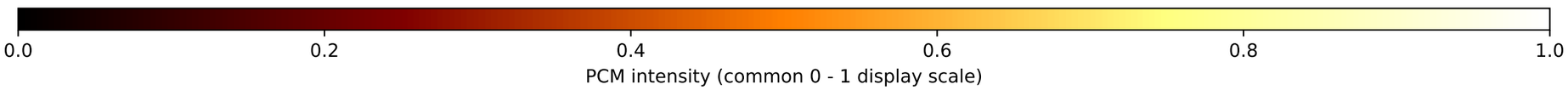
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   23.46 dB  0.411   0.365
10%   25.82 dB  0.448   0.396
20%   26.33 dB  0.452   0.409
30%   26.46 dB  0.449   0.398
40%   24.93 dB  0.428   0.359
50%   26.57 dB  0.459   0.402
```

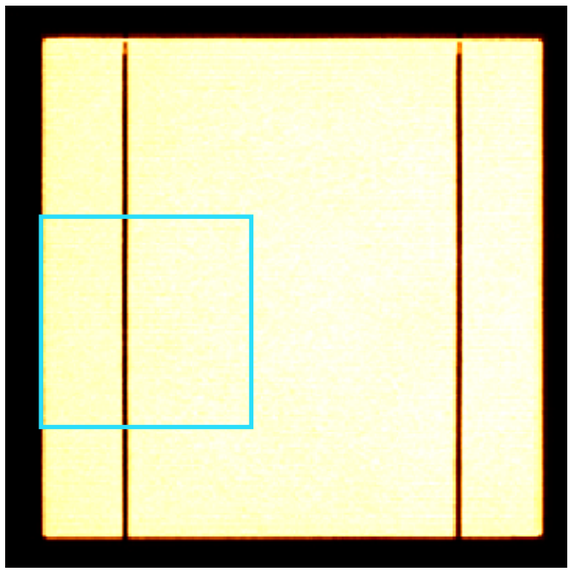
Crop: x=16:112, y=144:240. All panels use the same coordinates and intensity scale.



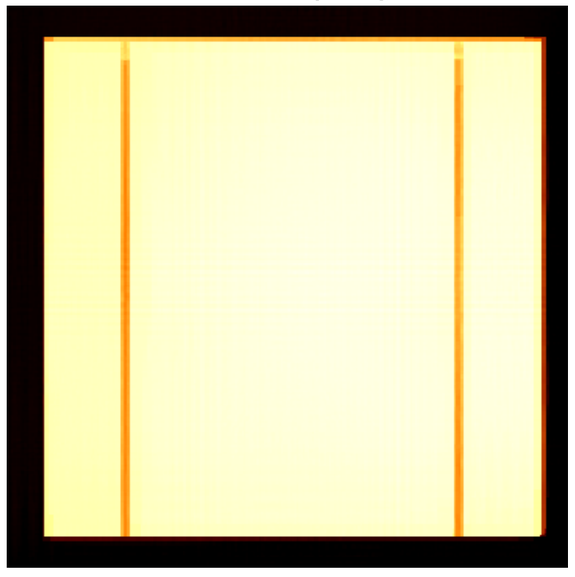
Silicone / image0249 | Full 256x256 field

HELD-OUT TEST SPLIT

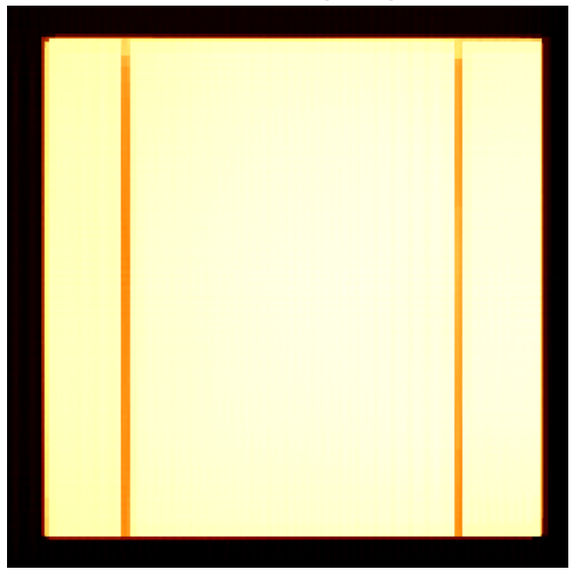
Ground truth



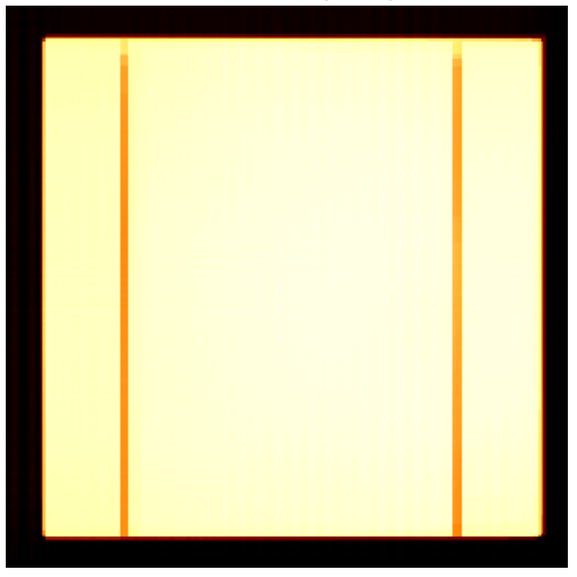
5% sampling



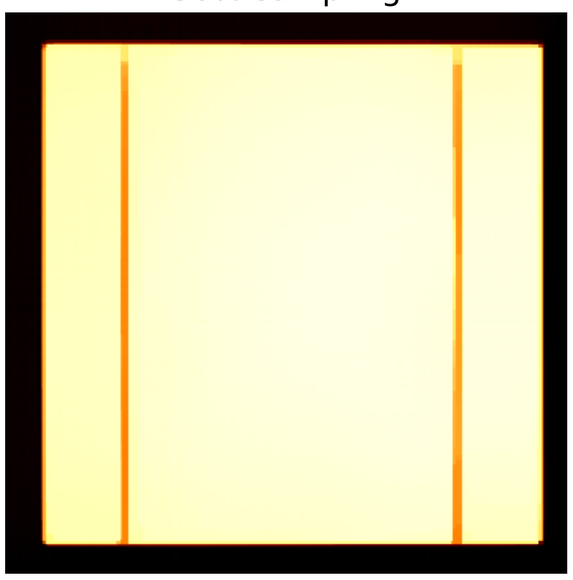
10% sampling



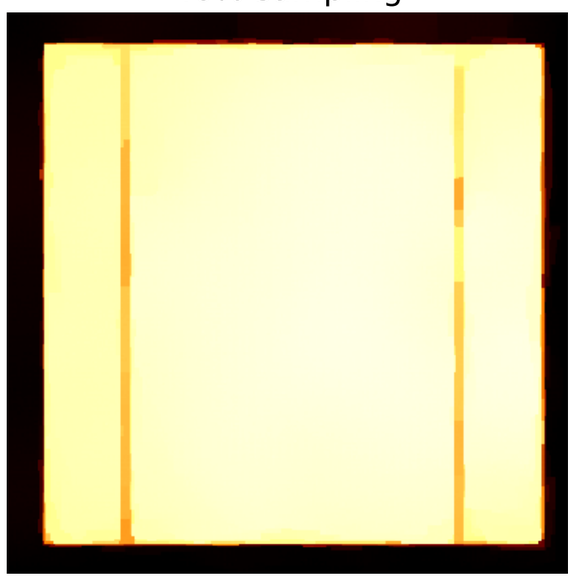
20% sampling



30% sampling



40% sampling



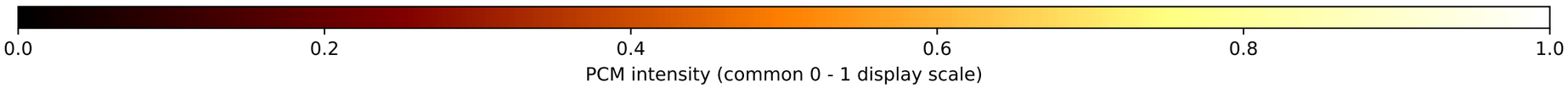
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

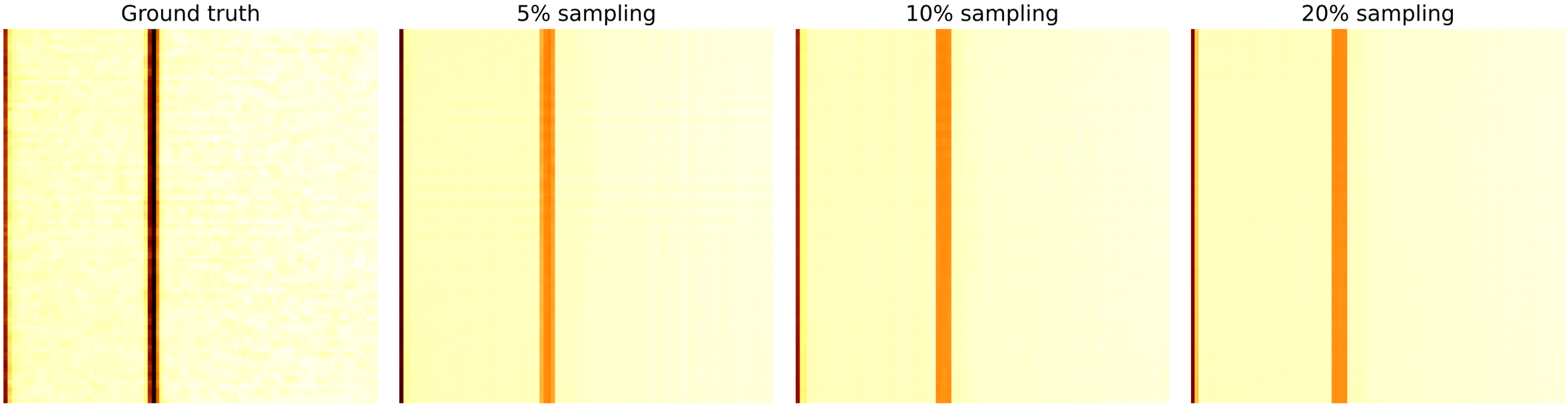
PSNR / SSIM / HaarPSI
 5%   24.16 dB  0.669   0.584
10%   25.08 dB  0.677   0.670
20%   25.24 dB  0.684   0.645
30%   25.85 dB  0.701   0.731
40%   23.45 dB  0.663   0.609
50%   20.28 dB  0.587   0.384
```

Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



Silicone / image0249 | Local detail (automatically selected gradient-rich 96px crop)

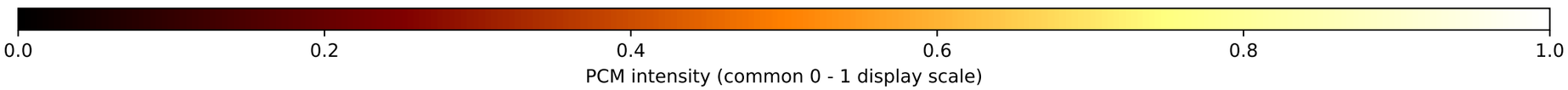
HELD-OUT TEST SPLIT




```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   24.16 dB  0.669   0.584
10%   25.08 dB  0.677   0.670
20%   25.24 dB  0.684   0.645
30%   25.85 dB  0.701   0.731
40%   23.45 dB  0.663   0.609
50%   20.28 dB  0.587   0.384
```

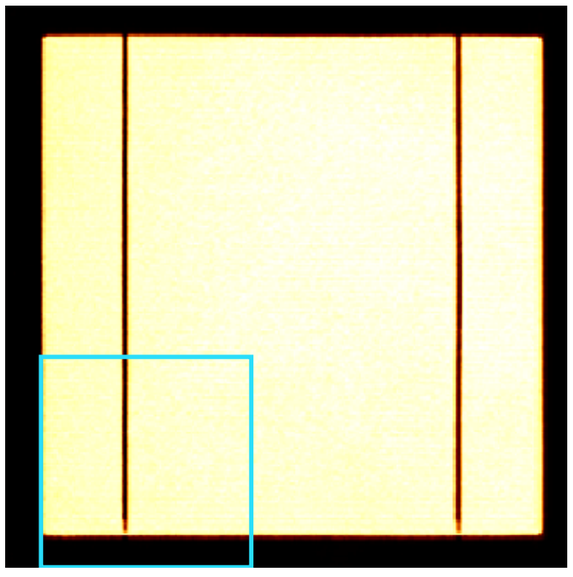
Crop: x=16:112, y=96:192. All panels use the same coordinates and intensity scale.



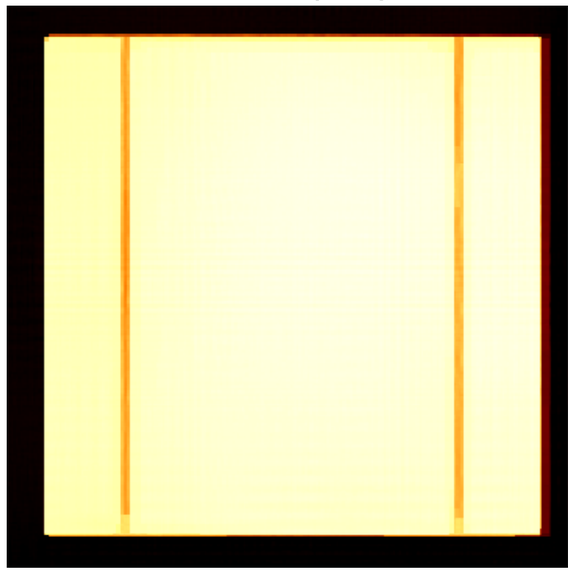
Silicone / image0288 | Full 256x256 field

HELD-OUT TEST SPLIT

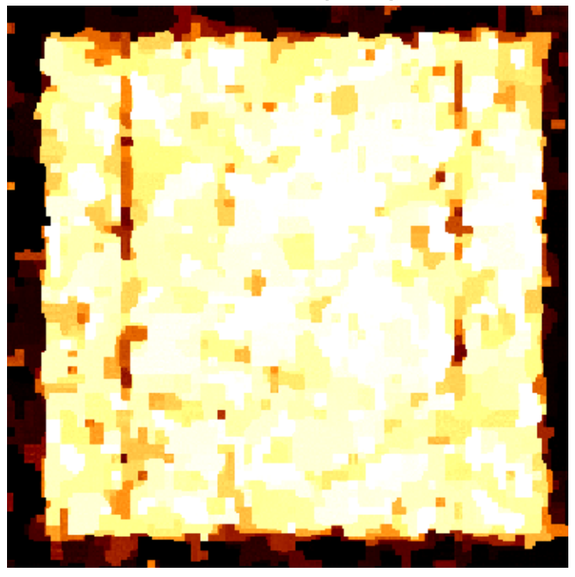
Ground truth



5% sampling



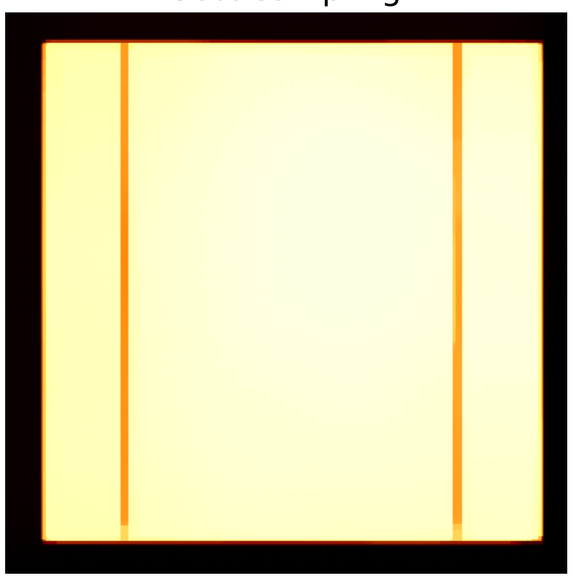
10% sampling



20% sampling



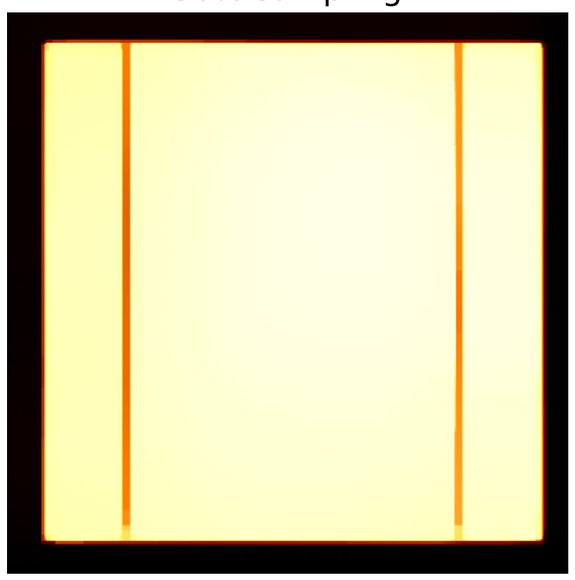
30% sampling



40% sampling



50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

	PSNR	SSIM	HaarPSI
5%	23.69 dB	0.661	0.588
10%	15.57 dB	0.273	0.234
20%	22.33 dB	0.629	0.515
30%	25.39 dB	0.698	0.736
40%	20.96 dB	0.618	0.418
50%	26.89 dB	0.713	0.784

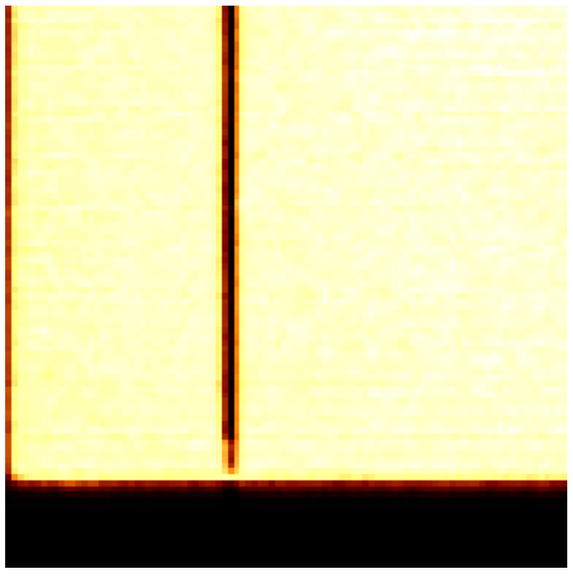
Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



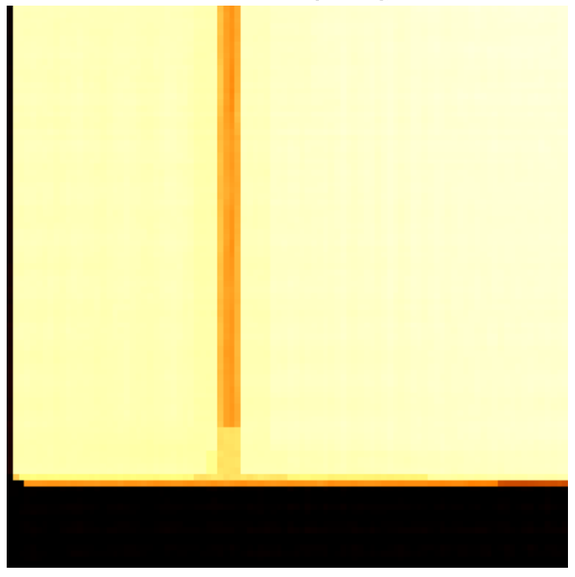
Silicone / image0288 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT

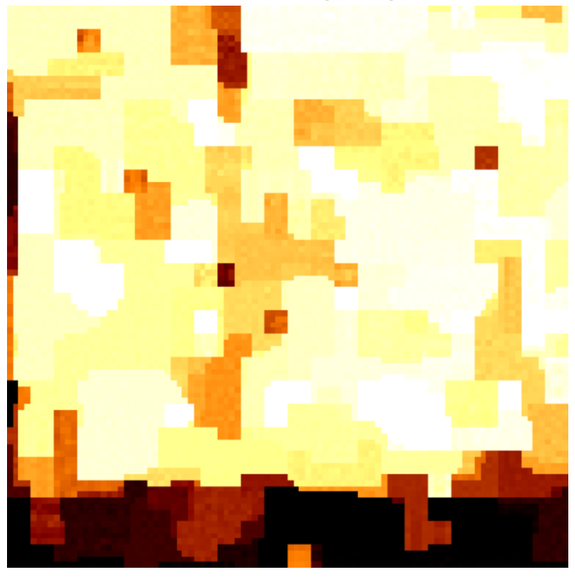
Ground truth



5% sampling



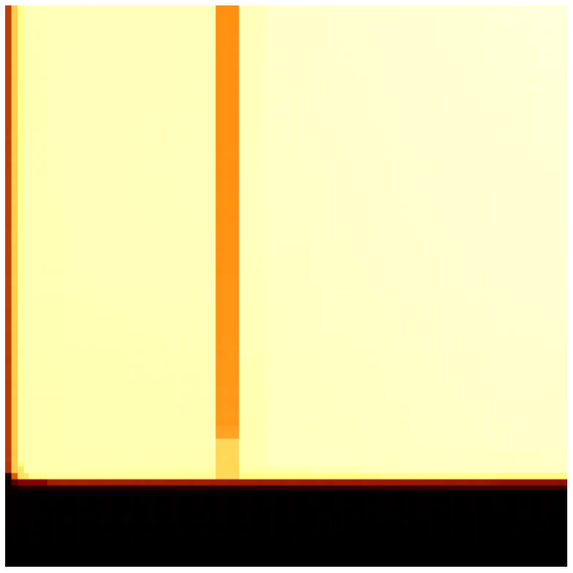
10% sampling



20% sampling



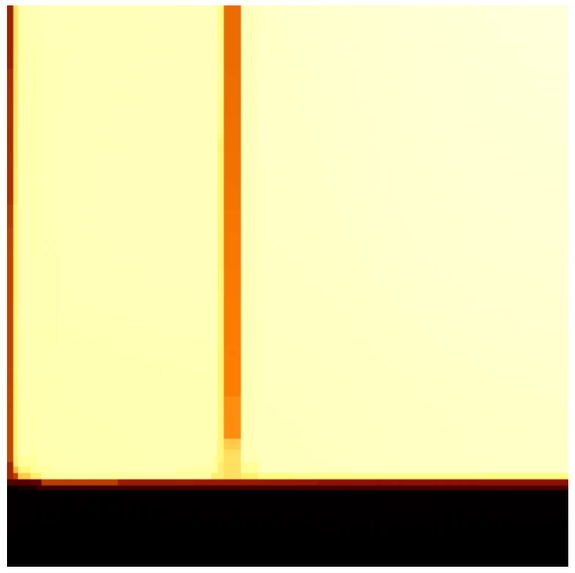
30% sampling



40% sampling



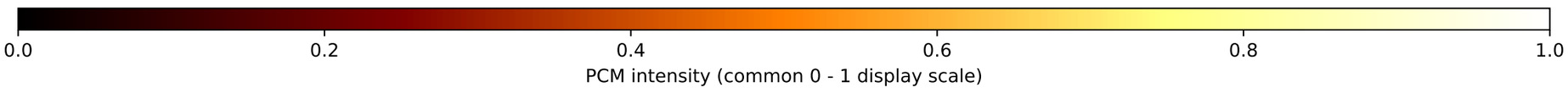
50% sampling



```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   23.69 dB  0.661   0.588
10%   15.57 dB  0.273   0.234
20%   22.33 dB  0.629   0.515
30%   25.39 dB  0.698   0.736
40%   20.96 dB  0.618   0.418
50%   26.89 dB  0.713   0.784
```

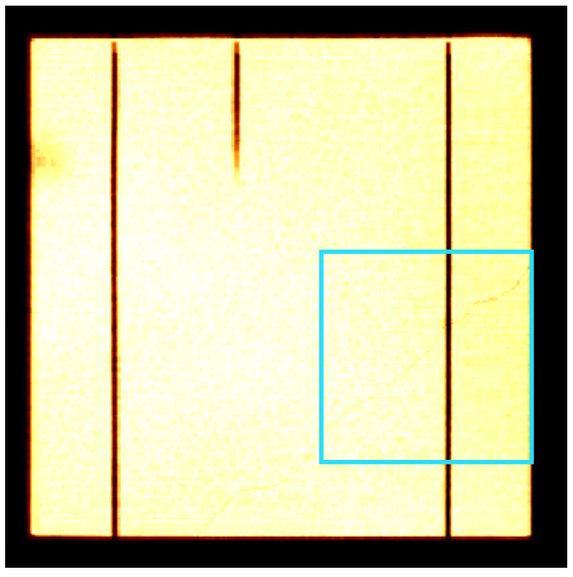
Crop: x=16:112, y=160:256. All panels use the same coordinates and intensity scale.



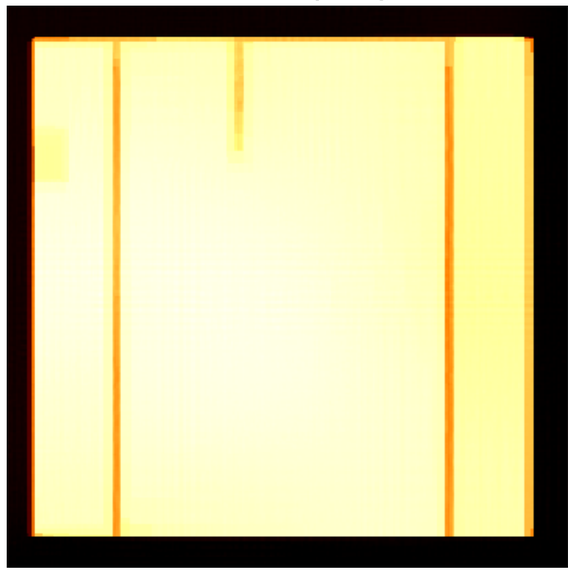
Silicone / image0611 | Full 256x256 field

HELD-OUT TEST SPLIT

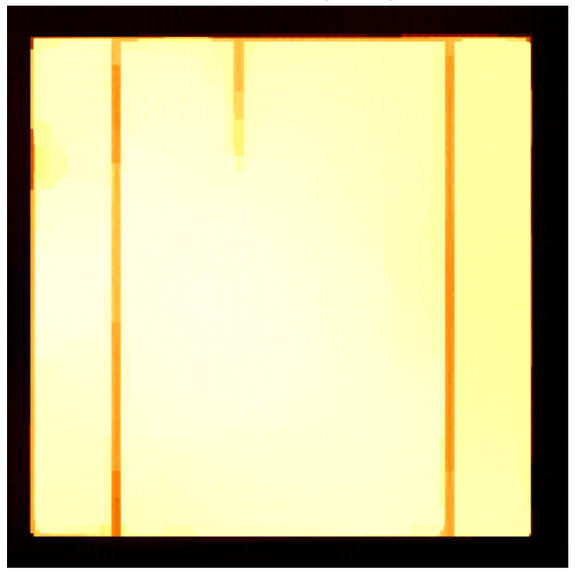
Ground truth



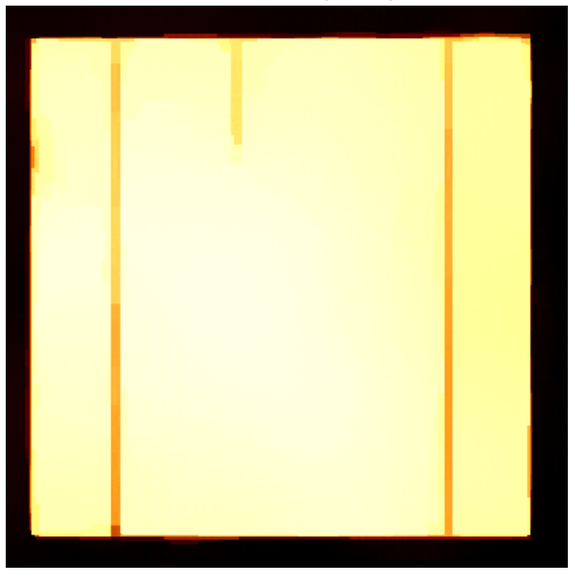
5% sampling



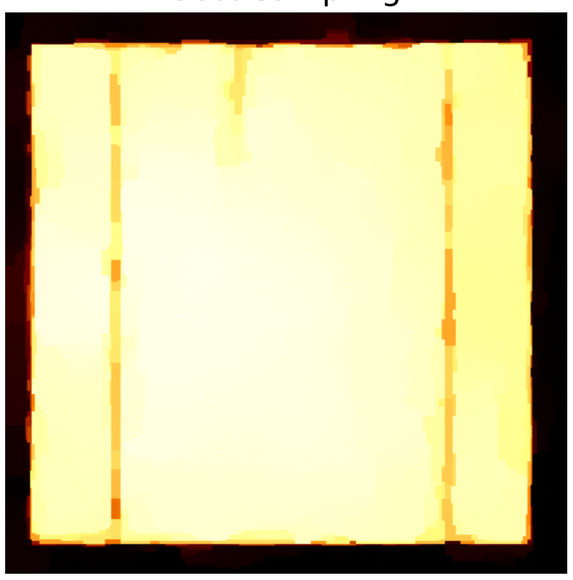
10% sampling



20% sampling



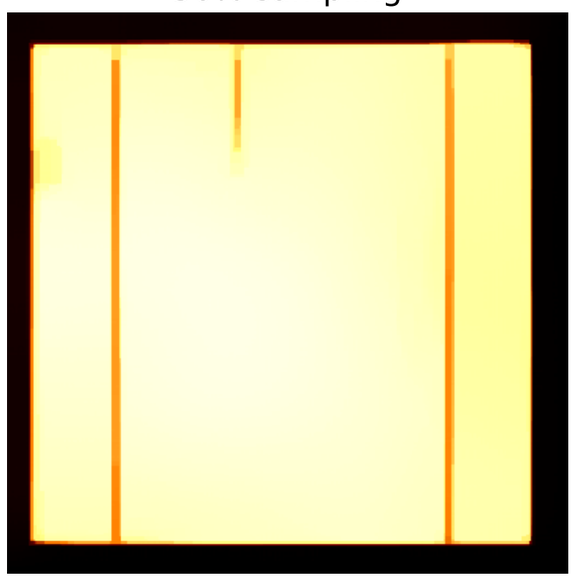
30% sampling



40% sampling



50% sampling



CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

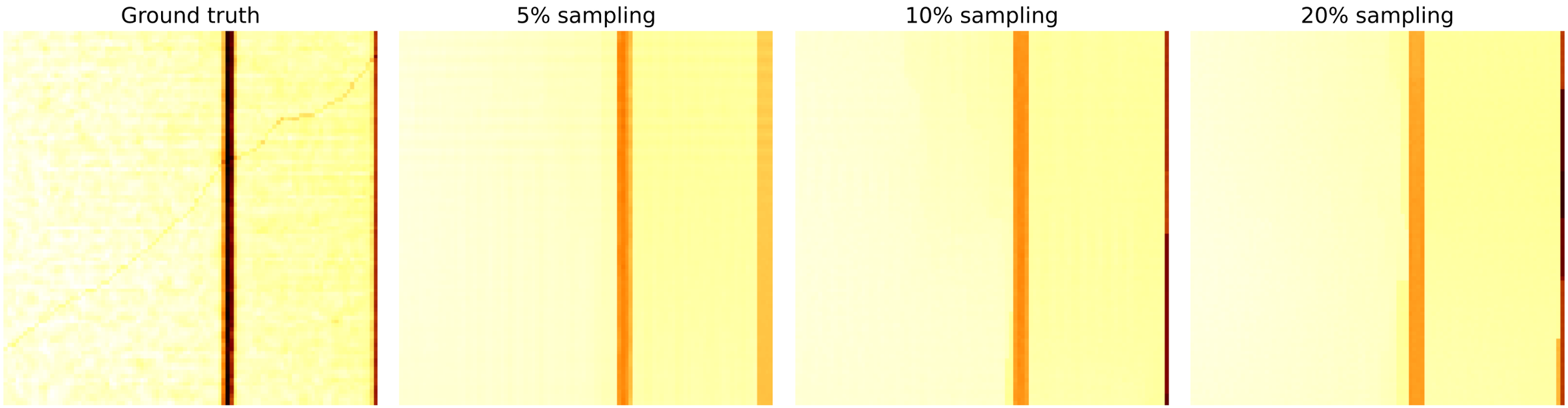
PSNR / SSIM / HaarPSI			
5%	23.05 dB	0.595	0.492
10%	23.97 dB	0.606	0.600
20%	23.59 dB	0.593	0.594
30%	22.16 dB	0.565	0.504
40%	18.76 dB	0.460	0.313
50%	25.35 dB	0.640	0.680

Cyan box indicates the crop shown on the following page. No per-image autoscaling is used.



Silicone / image0611 | Local detail (automatically selected gradient-rich 96px crop)

HELD-OUT TEST SPLIT




```
CHECKPOINT: epoch 5 / step 22500
Test ratios: 5, 10, 20, 30, 40, 50%
Image values: shared 0 - 1 display

PSNR / SSIM / HaarPSI
 5%   23.05 dB  0.595   0.492
10%   23.97 dB  0.606   0.600
20%   23.59 dB  0.593   0.594
30%   22.16 dB  0.565   0.504
40%   18.76 dB  0.460   0.313
50%   25.35 dB  0.640   0.680
```

Crop: x=144:240, y=112:208. All panels use the same coordinates and intensity scale.

